

Design and Development of low-cost bio inspired soft robot for pipeline inspection



EN8911 – ENGINEERING RESEARCH PROJECT

Submitted by

Fadhel Ali - 201902117

Supervised by

Dr. Belal ElSayed

Dr. Ahmed Abdulrahman

DEPARTMENT OF MECHANICAL ENGINEERING
SCHOOL OF ENGINEERING, BAHRAIN POLYTECHNIC
KINGDOM OF BAHRAIN

2025

AUTHOR'S DECLARATION

I hereby declare that I am the sole author of this thesis. It is further declared that I have fulfilled all the requirements in line with the Quality Assurance guidelines of the Bahrain Polytechnic.

I understand that my report may be made electronically available to the public.

Signature: 

Name: Fadhel Ali

Date: 5/25/2025

Abstract

This project presents the design, simulation, and fabrication of a low-cost, bio-inspired soft robot crawler intended for internal pipeline inspection. The crawler consists of two radial expansion actuators and a central linear actuator, all made from hyperelastic silicone (Ecoflex™ 00-10), allowing it to adapt to small-diameter and curved pipelines. CAD models were developed using SolidWorks and fabricated through 3D-printed PLA molds. Static structural analysis was performed in ANSYS using the Yeoh 3rd-order hyperelastic model to evaluate total deformation, elastic strain, and von Mises stress under 100 kPa of internal pressure.

The balloon component demonstrated safe and uniform radial expansion with a maximum strain of 62.8% and stress of 85.4 kPa, remaining well within material limits. The linear actuator achieved 174.36% elongation but exceeded its tensile strength with a stress peak of 15.81 MPa, indicating the need for material or pressure adjustments. The final design addressed the failures of early prototypes by improving wall thickness, geometry, and mold alignment. Results confirm the crawler's effectiveness for safe, adaptive locomotion inside pipelines, offering a scalable solution for energy infrastructure maintenance.

Acknowledgements

I would like to express my deepest gratitude to my supervisor, Dr. Belal ElSayed, for his continuous support, expert guidance, and encouragement throughout every stage of this project. His knowledge and insight were instrumental in shaping the outcome of this work.

I would also like to thank Dr. Ahmed Abdulrahman for his valuable advice and feedback, which helped improve the technical and academic quality of this Project.

Sincere appreciation goes to the Bahrain Polytechnic engineering faculty and technical staff for providing the tools, facilities, and support needed to complete the simulations, fabrication, and testing phases of this project. Their assistance was critical to bringing this research to life.

Table of Contents

1	Introduction.....	9
1.1	background	9
1.2	Problem statement.....	10
1.3	Project scope.....	11
1.4	Project objectives.....	12
1.5	Bahrain’s perspective	13
2	Literature review	14
2.1	Yu et al. (2023): Design and Motion Simulation of a Soft Robot for Crawling in Pipes	14
2.2	Nguyen et al. (2024): Soft Robot Employing a Series of Pneumatic Actuators and Distributed Balloons	15
2.3	Bar-Cohen (2014): Electroactive Polymers as Soft Robotic Actuators	16
3	Methodology.....	17
3.1	Overview of the Project.....	17
3.2	Project Management Framework	17
3.3	Design Process	21
3.3.1	Linear Actuator	21
3.3.2	Solid Radial Shell (Grooved Part)	22
3.3.3	Internal Balloon (Radial Grip Element)	23
3.3.4	Connector	24
3.4	CAD Modelling and Design Iteration.....	25
3.4.1	Linear Actuator (Design one)	26
3.4.2	Linear Actuator (Final design)	27
3.4.3	Radial actuator (Design one)	29
3.4.4	Radial Shell (Final design)	30
3.4.5	Connector Segment Modelling	32
3.4.6	Internal Balloon Modelling	33
3.5	Final Device Assembly Overview	34
3.6	Mold design	35

3.6.1	Linear Actuator Mold Design Iteration	35
3.6.2	Linear Actuator Mold Design Iteration	37
3.6.3	Radial Balloon Mold Design.....	38
3.6.4	Balloon seal mold	39
3.7	Fabrication of Molds using 3D Printers	40
3.7.1	Printer Setup.....	40
3.7.2	Fabrication Issues and Resolution	43
3.8	Simulation	45
3.8.1	Step 1: Material Assignment – Hyperelastic Yeoh Model for DoYB Silicone..	45
3.8.2	Step 2: Geometry Import and Internal Face Selection	47
3.8.3	Step 3: Assigning Material in Mechanical Model.....	49
3.8.4	Step 4: Mesh Settings and Element Control.....	50
3.8.5	Step 5: Static Structural Settings	52
3.8.6	Step 6: Solution Setup and Result Requests	56
3.8.7	Step 7: Final Mesh Generation and Solution	57
3.9	Simulation of Balloon Expansion in ANSYS	58
3.9.1	Static Structural Study Creation	58
3.9.2	Material Assignment – ECOFLEX00-10 Silicone.....	58
3.9.3	Geometry Import.....	59
3.9.4	Mesh Generation and Element Setup.....	59
3.9.5	Boundary Conditions.....	60
3.9.6	Pressure Loading.....	61
3.9.7	Run simulation.....	61
4	Results.....	62
4.1	Early Design Limitations (Linear)	62
4.2	Finalized Linear Actuator design/model	64
4.2.1	Deformation calculation	65
4.2.2	Material strain data.....	67
4.2.3	Equivalent stress results	69

4.3	Total Deformation – Balloon Simulation	71
4.3.1	Equivalent Elastic Strain – Balloon Simulation	75
4.3.2	Equivalent Stress (von Mises) – Balloon Simulation	78
5	Discussion of Results	81
6	Conclusion	82
7	References	83

1 Introduction

1.1 background

Pipelines are a critical part of modern infrastructure, transporting oil, gas, water, and industrial fluids. Over time, these pipelines face degradation due to corrosion, blockages, or material fatigue leading to costly maintenance, environmental hazards, and operational downtime. Traditional inspection methods, often relying on rigid robots or manual access, are limited in their ability to safely and efficiently navigate confined or curved pipeline environments.

Soft robotics presents a promising alternative, using flexible, compatible materials to achieve motion and adaptability being bio inspired by nature's earthworm. These systems offer lower risk of internal pipe damage and better navigation through variable geometries. This project introduces a low-cost, bio-inspired soft robot crawler designed to perform pipeline inspection using pneumatic soft actuators for both radial gripping and linear locomotion. The robot is specifically designed to address industry challenges in Bahrain and align with sustainable maintenance practices.

1.2 Problem statement

Most pipeline inspection robots in use today are rigid, complex, and limited to large-diameter pipelines. For instance, the commercially available FW100CT.2 robot is only compatible with pipes of 100 mm and above, making it unsuitable for narrower or more variable pipe systems commonly found in localized or aging infrastructure (RICO GmbH, n.d.). While the FW100CT.2 features advanced capabilities such as steerable wheels and integrated sensors, its size, complexity, and cost make it less accessible for low-budget operations or routine inspections. Furthermore, as highlighted by PipeDetect (n.d.), customers often face additional challenges with traditional robots, including high maintenance frequency, limited-service life, and complex operating procedures that require skilled users. These issues reduce efficiency, increase operational downtime, and make current solutions impractical for environments that require flexibility and low-cost implementation. Therefore, there is a clear need for a lightweight, adaptable, and damage-free inspection solution that can operate in smaller pipes (as small as 4 inches or 101.6 mm), requires minimal maintenance, and is easy to control. The proposed soft robot crawler addresses these needs using flexible, bio-inspired materials and pneumatic soft actuators, offering a non-disruptive, affordable, and user-friendly alternative to traditional rigid systems.

In addition to the limitations of current technologies, many pipeline inspections, especially in the oil and gas sector are still performed manually, exposing human workers to confined, hazardous environments. This method exposes workers to serious safety and health hazards, and in the unfortunate event of an accident, companies face severe consequences. These include not only the potential loss of life but also legal liabilities and substantial insurance payouts. According to global industry trends, workplace incidents in pipeline systems can cost oil companies millions in legal compensation and operational downtime, significantly impacting their financial and reputational standing (API, 2021). Furthermore, a detailed study by the Interstate Natural Gas Association of America (INGAA) found that pipeline testing and inspection costs can range from \$200 to over \$417 per linear foot, depending on the pipeline's location and risk classification contributing to an estimated \$1.3 billion annually (INGAA, 2016). These figures emphasize the urgent need for safer, automated, and cost-effective alternatives that minimize human exposure while maintaining inspection accuracy and regulatory compliance.

1.3 Project scope

This project focuses on the design, simulation, and analysis of a soft robot crawler for internal pipeline inspection. The robot's earthworm-like motion is bio-inspired. It utilizes pneumatic soft actuators to achieve both radial expansion for gripping the pipe walls and linear extension for forward crawling.

The scope of the project includes:

- CAD modeling of the robot and its actuator components using SolidWorks
- Hyperelastic material simulation using ANSYS to analyze deformation, strain, and stress behavior
- Material selection for actuator casting based on flexibility, durability, and real-world performance
- Prototype development

1.4 Project objectives

The primary objective of this project is to design and analyze a soft robot crawler capable of performing safe, flexible, and efficient pipeline inspections using soft pneumatic actuators. The specific objectives are:

- To study the limitations of existing pipeline inspection methods and evaluate the potential of soft robotics for confined environments.
- To design a soft robot crawler model featuring radial and linear expansion mechanisms, inspired by earthworm-like locomotion.
- To develop detailed 3D CAD models of the robot using SolidWorks, focusing on actuator geometry and integration.
- To simulate the actuator's mechanical behavior using ANSYS, focusing on total deformation, elastic strain, and stress using hyperelastic analysis.
- To prototype selected parts (such as the linear actuator and radial balloon mechanism) using soft silicone casting.
- To assess the deformation performance and suitability of materials under pressure for crawling movement within 3-inch pipelines.

1.5 Bahrain's perspective

Bahrain's economy is heavily reliant on its oil and gas sector, with companies such as Bapco Energies operating massive underground pipeline networks across the Kingdom. These pipelines are vital for the transportation of crude oil and refined products, and their maintenance is essential to avoid leaks, corrosion, or costly downtime. Traditional pipeline inspection methods in Bahrain often involve manual labor or rigid robotic systems, which are limited in their adaptability and pose significant risks in narrow or aging pipelines. In the event of inspection failures or accidents, companies not only face operational delays but also high compensation and insurance liabilities.

The development of a low-cost soft robot crawler addresses a direct need in Bahrain's industrial context. Designed to navigate pipes as narrow as 3 inches, this soft robot offers a safe, flexible, and efficient alternative for internal inspection, especially in environments where traditional robots cannot operate. Its bio-inspired earthworm motion and use of soft pneumatic actuators ensure minimal damage to internal pipe walls and eliminate the need for human entry into confined, hazardous areas.

Moreover, this project aligns with Bahrain's Economic Vision 2030, which emphasizes sustainable development, innovation, and a shift toward a knowledge-based economy. It also pairs with Bapco's recent initiatives aimed at modernizing pipeline infrastructure, such as the new AB-4 pipeline project connecting Bahrain to Saudi Arabia. The proposed crawler could eventually support routine inspection, enhancing operational reliability while reducing long-term maintenance costs.

2 Literature review

2.1 Yu et al. (2023): Design and Motion Simulation of a Soft Robot for Crawling in Pipes

Yu et al. (2023) presented a soft robot specifically designed for pipe crawling, incorporating both bending and elongation capabilities through modular actuator segmentation. The robot structure includes two deformable ends (front and rear) and a central expansion section, enabling inchworm-like locomotion. Each segment utilizes pneumatic actuation to either grip the pipe walls or extend the body, allowing the robot to crawl through confined spaces. The design was simulated using ABAQUS to test locomotion within pipelines of 20 mm to 42.4 mm diameter, including horizontal and curved sections.

The robot body was fabricated using silicone rubber and casted in mold assemblies, supporting soft behavior while maintaining resilience. The experimental validation confirmed smooth motion and strong adaptability in bent pipe environments. This study provides critical insights into this project's design of radial and linear actuators, particularly the integration of gripping and crawling. The performance validation via simulation also aligns with this project's approach, where hyperelastic material simulations are conducted in ANSYS to confirm deformation characteristics.

Key Contribution to Current Work:

- Validates the use of pneumatic chamber-based motion.
- Confirms soft robots' effectiveness in navigating tight, curved pipelines.
- Reinforces the feasibility of inchworm-like movement for confined inspection.

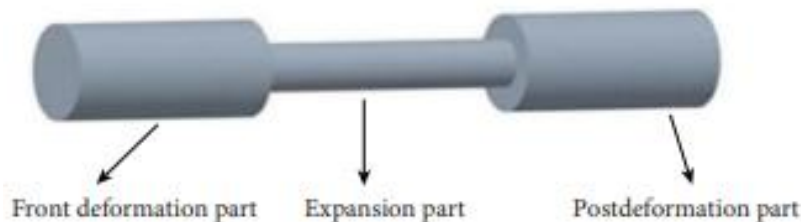


FIGURE 1: Schematic diagram of the pipe soft robot.

Figure 1 2.1 Yu et al. (2023)

2.2 Nguyen et al. (2024): Soft Robot Employing a Series of Pneumatic Actuators and Distributed Balloons

Nguyen et al. (2024) developed a soft pipeline inspection robot based on serial pneumatic actuators with embedded distributed balloon structures. These balloons serve as contact nodes, anchoring the robot against pipe walls during movement and enabling wave-like locomotion. The system operates via alternating pressure cycles, causing the actuator to steadily inflate and deflate along its length. The robot achieved a maximum forward velocity of 40.11 mm/s in horizontal settings and 9.22 mm/s in vertical climbing tests, showing high efficiency and controllability.

A key feature of the design is the modular configuration, allowing easy replication of actuator units for different lengths or strength requirements. The research also includes a theoretical model that estimates output force and motion velocity, verified through experimental data. This directly supports the development of the balloon-based radial actuator in this project, where internal inflation through radial grooves allows temporary anchoring against pipe walls.

Key Contribution to Current Work:

- Demonstrates effectiveness of balloon-based soft gripping systems.
- Offers a performance benchmark for radial actuator deformation and crawling speed.
- Supports modular actuator modeling and force estimation.

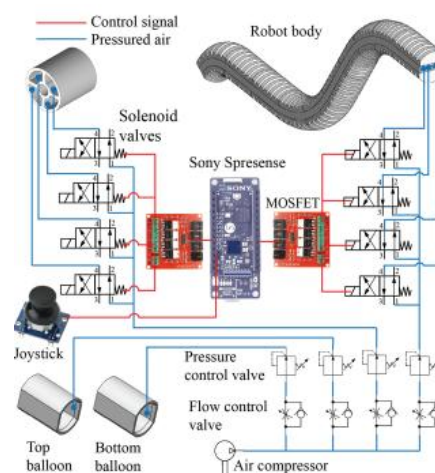


Figure 2.2 Nguyen et al. (2024):

2.3 Bar-Cohen (2014): Electroactive Polymers as Soft Robotic Actuators

Bar-Cohen (2014) provides a comprehensive review of electroactive polymers (EAPs) as a category of smart materials capable of producing large and flexible deformation under electrical stimulation. While the focus is not fully on pipeline inspection, the paper outlines the principles of soft actuator behavior, including energy efficiency, mechanical responsiveness, and material limitations. EAPs are shown to demonstrate muscle-like contractions and are proposed for applications requiring low weight, silent operation, and adaptability characteristics needed in soft pipeline robots.

Although the current project does not use EAPs directly (opting for pneumatic silicone-based systems), the study offers valuable context on material selection criteria, actuation behavior under stress, and the importance of safety and compliance in sensitive environments. It reinforces the need for low-voltage, high-deformation materials, which this project addresses through silicone actuators and pneumatic inflation.

Key Contribution to Current Work:

- Establishes soft actuation design principles.
- Highlights the importance of safe, flexible material deformation.
- Provides a broader context for selecting non-metallic actuator materials.

3 Methodology

3.1 Overview of the Project

This project follows a structured engineering design methodology that integrates conceptual development, digital modeling, simulation-based validation, and physical prototyping. The primary goal is to design a soft robot crawler that can navigate pipelines using bio-inspired earthworm locomotion, achieved through the expansion of radial and linear soft actuators. The project emphasizes a low-cost, modular design using silicone materials and 3D-printed molds to ensure manufacturability and adaptability in real pipeline conditions.

3.2 Project Management Framework

The following Gantt chart outlines the structured and phase-wise development of the soft robot crawler from initiation to closure. Each phase was meticulously planned to ensure a smooth transition from literature review to CAD design, simulation, prototyping, and testing.

Project Initiation/Planning

- Conducted weekly meetings with the supervisor to clarify project goals and refine direction.
- Reviewed academic literature related to pipeline inspection and soft robotics, particularly focusing on pneumatic actuation and earthworm-inspired locomotion.
- Identified limitations of current rigid robots and defined technical goals aligned with Bahrain's pipeline inspection needs.
- Established the initial timeline, responsibilities, and budget structure.

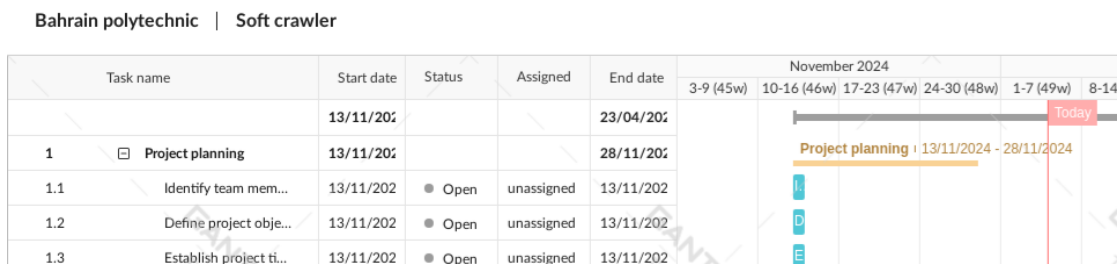


Figure 3.2 Planning Phase

Research and development

- Gathering academic sources related to soft robotics and pipeline inspection
- Studying soft actuator designs (radial and linear)
- Reviewing materials suitable for soft actuators (e.g., Ecoflex, DoYB)
- Analyzing previous pipeline robot locomotion methods
- Compiling relevant data and extracting design insights

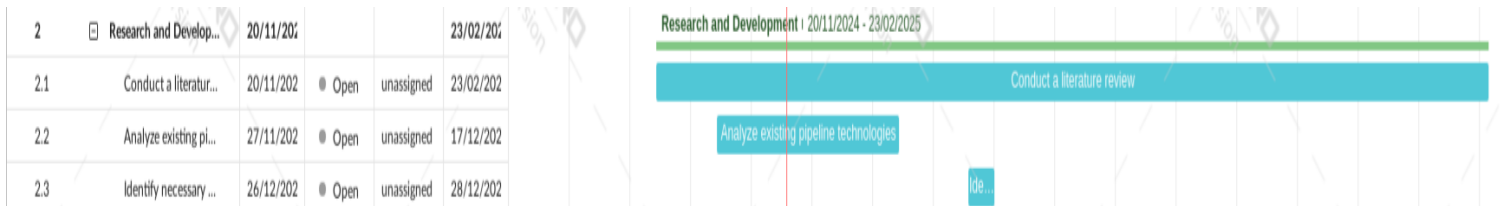


Figure 3.2 Research and development Phase

Design Phase

- Creating SolidWorks CAD models of:
 - Radial actuator with balloon cavity
 - Central linear actuator
 - Mold assemblies for silicone casting
- Performing dimensional analysis for pipe fitment (target diameter: 4 inches / 101.6 mm)



Figure 3.2 Design Phase

Prototyping and Fabrication

- Printing actuator molds using generic PLA filament
- Mixing silicone materials (A:B ratio 1:1)
- Pouring, curing, and demolding of soft actuators
- Embedding balloon structures into radial sections
- Final assembly of robot module with connectors



Figure 3.2 Prototyping & Fabrication Phase

Software and Simulation

- Exporting geometry from SolidWorks to ANSYS
- Assigning Yeoh hyperelastic material properties
- Setting up boundary conditions for pressure analysis
- Running structural simulations to test:
 - Total deformation (achieved: 19.66% stretch)
 - Elastic strain and Von Mises stress distribution



Figure 3.2 Simulation Phase

Testing and Evaluation

- Pressurizing linear and radial actuators in controlled tests
- Measuring expansion, contraction, and grip strength
- Observing stability
- Comparing test data with simulation results

Testing Phase	02/02/2025			07/02/2025
Perform initial asse...	02/02/2025	● Open	unassigned	03/02/2025
Gather feedback fr...	03/02/2025	● Open	unassigned	05/02/2025
Analyze test results ...	05/02/2025	● Open	unassigned	07/02/2025



Figure 3.2 Testing Phase

Field Adjustment

- Making design refinements based on testing issues
- Adjusting balloon thickness and silicone layering
- Minor reprinting of mold parts for improved fit

Field testing	08/02/2025			10/02/2025
Plan field test in co...	08/02/2025	● Open	unassigned	10/02/2025
Execute test in an a...	08/02/2025	● Open	unassigned	10/02/2025



Figure 3.2 Field Adjustment Phase

Documentation and closing of project

- Writing final thesis chapters and project report
- Preparing graduation expo presentation materials
- Creating posters, labels, and documentation for exhibition
- Final internal review with supervisor

Documentation and re...	16/02/2025			23/02/2025
Document each pr...	16/02/2025	● Open	unassigned	23/02/2025
Project review	13/11/2024			23/04/2025
Discuss outcomes	24/02/2025	● Open	unassigned	25/02/2025
Gather feedback on	25/02/2025	● Open	unassigned	28/02/2025



Figure 3.2 Project Closing Phase

3.3 Design Process

The design of the soft robot crawler is based on earthworm-like locomotion, achieved by alternating between gripping and expanding phases using soft actuators. The robot includes three main parts: two radial anchoring actuators (front and rear) and a central linear expansion actuator. The radial sections feature a solid grooved shell housing an internal balloon, which inflates outward to grip the pipe walls. The linear section expands lengthwise to push the body forward when the radial parts are fixed.

3.3.1 Linear Actuator

The linear actuator is the primary locomotion element responsible for forcing the crawler forward. It is located between the two radial actuators and operates via pneumatic inflation.

Design Features:

- Shape: Rectangular, bellows-like geometry with internal air chamber
- Material: Soft silicone (DoYB) to allow high elasticity
- Motion Type: Axial (longitudinal) extension upon pressurization
- Connection: Molded with inlet port for tubing; attached to radial ends via silicone bonding

Function:

Upon inflation, the actuator elongates in the axial direction, pushing the front segment forward while the rear radial actuator remains anchored. This mimics the stretch phase of an inchworm. The actuator returns to its original shape when depressurized, allowing reset for the next cycle.

Constraints:

- Must produce enough extension force to overcome friction and pipeline resistance
- Maximum pressure must remain within elastic deformation range

3.3.2 Solid Radial Shell (Grooved Part)

This part acts as the anchoring frame for each end of the robot and is designed to house and control the deformation of the internal balloon.

Design Features:

- Material: PLA (3D printed) for rigid structure
- Shape: Cylindrical with internal cavity and external grooves
- Grooves: Slotted regions to allow balloon to extrude outward

Function:

When the balloon inside the shell is inflated, it expands through the grooves, pressing outward against the inner walls of the pipe. This secures the robot in place during linear actuator movement. When deflated, the balloon retracts, releasing the grip.

Design Considerations:

- Groove depth and spacing optimized to allow uniform balloon extrusion
- Shell rigidity ensures that expansion is directed only through grooves, not circumferentially
- Pipe diameter compatibility: Designed to grip 3-inch internal diameter

3.3.3 Internal Balloon (Radial Grip Element)

The balloon is a soft, inflatable component placed inside the radial shell, designed to anchor the crawler during operation.

Design Features:

- Material: Ultra-soft silicone (Ecoflex 00-10)
- Shape: Thin-walled cylindrical bladder (custom cast)
- Inflation: Pneumatic input via small tubing from external or onboard source
- Bonding: Loosely bonded inside the shell to allow free expansion

Function:

Upon pressurization, the balloon inflates outward through the grooves of the shell, gripping the internal surface of the pipe without damaging it. It acts like the foot of an earthworm, holding the crawler steady while the linear segment extends or contracts.

Constraints:

- Must expand uniformly through all grooves to ensure stable anchoring
- Must avoid overexpansion that could rupture the balloon or damage the shell

3.3.4 Connector

The connector serves as the mechanical interface between the solid radial grooved actuator shell and the central linear actuator. Its role is to secure both segments, ensure smooth pressure transmission, and maintain axial alignment of the robot.

Design Features:

- Material: Same silicone as the linear actuator (DoYB), allowing slight flexibility and bonding compatibility
- Shape: Cylindrical adapter
- Geometry: Outer diameter matches the linear shell, while the inner end embeds into the radial grooved part
- Embedded Tube Path: Channels or grooves should be added to guide air tubing without interference

Function:

- Ensures that the device is properly connected as a whole, both linear and radial parts
- Provides structural continuity between the soft and rigid sections

Bonding & Assembly:

- The connector is glued using the same material of the linear actuator (DoYB) between the flat face of the linear actuator and the PLA radial shell
- It may be designed with an overlap lip or snap-fit geometry to provide a lock, depending on tolerances

3.4 CAD Modelling and Design Iteration

All components of the soft robot crawler were designed using SolidWorks 2023, following modular design principles and soft-material manufacturing constraints. Each part was modeled individually with dimensional considerations that matched the internal diameter of a standard 2-3 inch pipeline. The goal was to create geometry that could both perform the intended locomotion function and be easily fabricated using 3D printing and silicone casting.

3.4.1 Linear Actuator (Design one)

Multiple versions were developed during the design phase to evaluate geometric performance, compatibility with the radial segments, and manufacturability. The earliest version, though structurally sound, had several mechanical limitations.

Table 3.41 Linear Actuator (Initial Design)

Design Parameter	Value
Total actuator length	100 mm
Number of bellows	10
Length per bellow segment	10 mm
Outer diameter of each bellow	25 mm
Wall thickness	1 mm
Air inlet port diameter	2 mm

This version consisted of 10 packed bellows, designed to inflate along the axial direction. While it demonstrated basic soft actuation, it proved inadequate in real-world trials.

Observed Issues:

- **Undersized Bellows:** The 25 mm outer diameter provided insufficient air chamber volume, limiting expansion force.
- **Low Pressure Capacity:** The 1 mm thin walls, while flexible, were prone to deformation and could not withstand moderate internal pressure without ‘breaking’
- **Limited Axial Displacement:** Even at full inflation, the actuator offered minimal forward push, insufficient for overcoming pipe friction.
- **Small Air Inlet:** The 2 mm port size restricted airflow, leading to slow response times and pressure lag during inflation cycles.

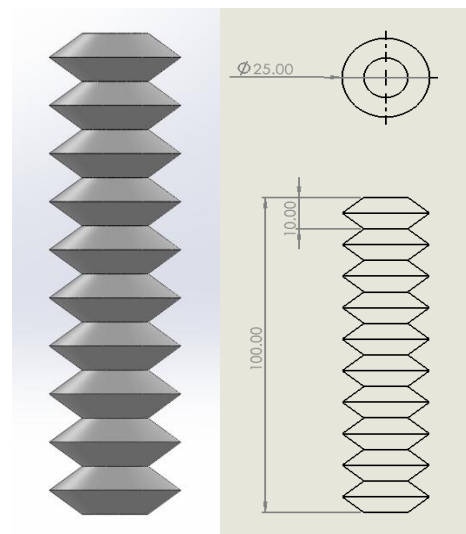


Figure 3.4.1 Initial Linear Design

3.4.2 Linear Actuator (Final design)

The linear actuator forms the central body of the soft robot crawler and is responsible for generating axial movement by inflating along its length. It was designed in SolidWorks 2023 using a bellows-like geometry to enable controlled extension under pneumatic pressure.

1. The design began with a 2D sketch of a single bellow unit.
2. This profile was then revolved using the *Revolve Boss/Base* feature.
3. The resulting geometry was patterned linearly to form a complete actuator composed of multiple bellows segments, 9 to be exact.
4. Each bellow was designed with an outer diameter of approximately 55 mm, ensuring sufficient material flex.
5. The wall thickness was maintained at 1.5 mm using the *offset* feature.
6. A small air inlet port (5 mm internal diameter) was added using *Extruded Boss*, going through the actuator body.

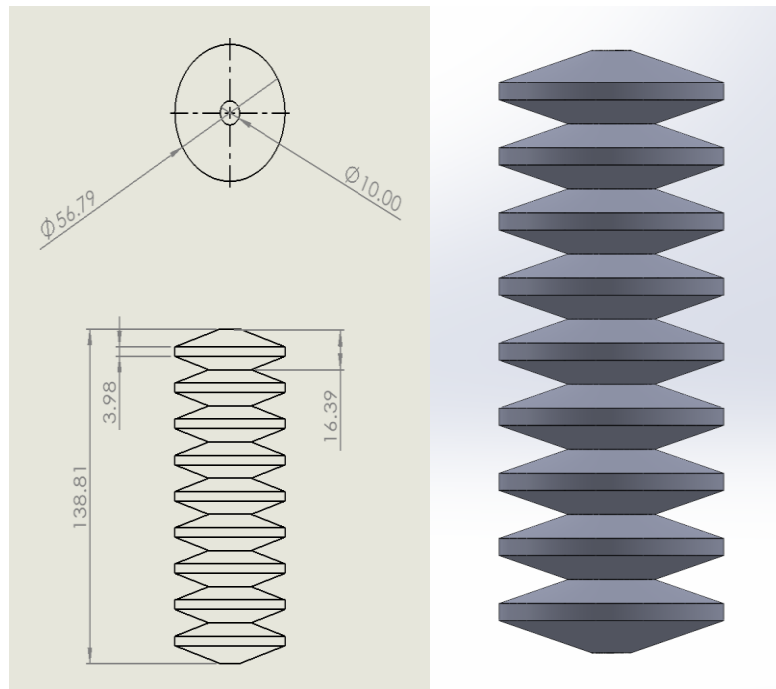


Figure 3.4.2 Linear Actuator Final Design

Final Design Enhancements

To resolve these issues, the final version of the linear actuator was significantly redesigned:

- Bellow diameter increased to 55 mm, matching pipe dimensions and increasing pneumatic capacity.
- Smoothed and filleted (2 mm) to reduce strain.
- Wall thickness was increased to 2 mm.
- Air inlet size increased in prototypes for testing, improving response speed and internal airflow control.
- Overall length extended to ~130 mm for enhanced stroke output.

3.4.3 Radial actuator (Design one)

The radial shell is a rigid cylindrical housing designed to anchor the robot to the pipe wall by channeling the expansion of an internal balloon. While the final design successfully balances strength, control, and modularity, the early design concept took a different approach that later proved structurally flawed.

Initial Design Concept – Fully Soft and Glued Assembly

Geometric Dimensions: Same as final version

- Outer diameter: 52 mm
- Inner diameter: ~48 mm
- Height: 50 mm
- Wall thickness: 4 mm
- Groove design: Same profile, created for balloon expansion
- Connection strategy: All components (radial, linear, and connectors) to be made of the same silicone material (Ecoflex or DoYB) and bonded together using silicone adhesive or by direct casting into each other.

Observed Issues:

- Weak Structural Integrity: Bonding soft silicone to silicone proved unreliable. Under pneumatic pressure, especially when both the linear actuator and balloon inflated simultaneously, the glued joints were expected to deform, peel, or rupture.
- No Mechanical Interlocking: Without a rigid connector or snap-fit system, there was no structural locking between components.
- Increased Rupture Risk: The soft on soft interfaces acted as failure points during the design and brainstorming period.
- Assembly Imbalance: Aligning and securing soft components was time-consuming and imprecise.

3.4.4 Radial Shell (Final design)

The radial shell was designed as a rigid cylindrical component that houses the internal balloon and directs its expansion outward through specially designed grooves. This part serves as the robot's anchoring mechanism, gripping the pipe wall during locomotion.

- The shell was modeled using the Extruded Boss/Base feature, with an outer diameter of approximately 52 mm and a total height (extrusion) of 50 mm.
- The inner diameter was set to approximately 48 mm, leaving a uniform wall thickness of 4 mm, providing structural rigidity while keeping the part compact enough to function within a 3-inch pipe.
- A 2D sketch of grooves was applied using the Wrap feature. These grooves, typically 4mm deep and 40mm wide, were evenly wrapped around the cylindrical surface to direct the expansion of the balloon.
- Two circular cuts (holes) were included at each end of the shell:
 - One side was designed with a 21 mm diameter hole to allow interfacing with the silicone connector and to secure the actuator assembly.
 - The opposite side featured a 7 mm hole, serving as a pass-through port for the pressurized air hose that inflates the balloon.

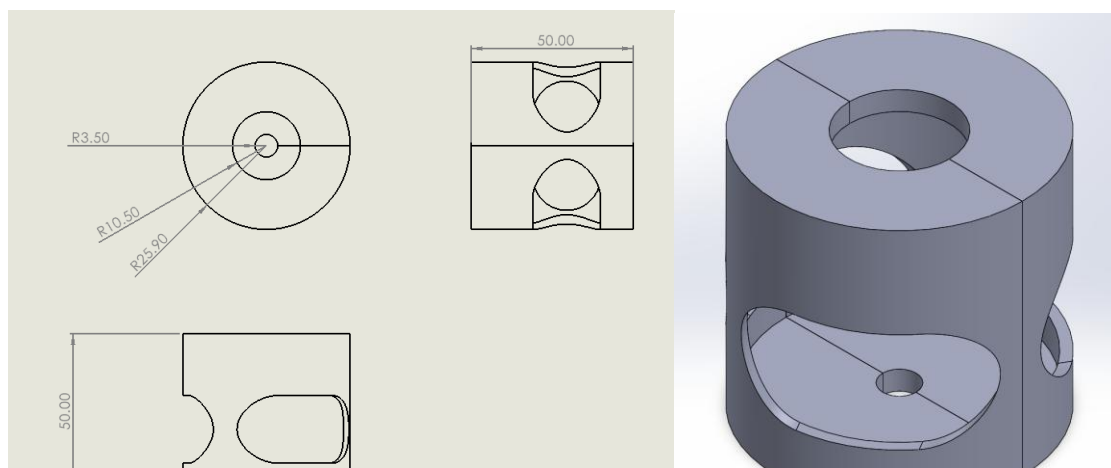


Figure 3.4.4 Final Shell Design

Final Design Enhancements

To overcome these limitations, the radial shell was re-engineered using a rigid modular approach:

- The shell was 3D printed from PLA, offering high structural rigidity to contain the balloon's expansion precisely through the grooves.
- The radial segment was designed to interface mechanically with the connector through a 45.79 mm socket, rather than relying on soft-to-soft adhesion.
- The updated design introduced two end holes:
 - A 21 mm opening to seat the silicone connector
 - A 5 mm air passage hole for balloon inflation

The rigid shell ensures all balloon pressure is translated outward rather than into the walls of adjacent parts, and the connector provides structural integrity, making the crawler more robust under repeated use.

3.4.5 Connector Segment Modelling

The connector segment is a designed interface that connects the linear actuator to the radial shell, ensuring axial alignment, sealed air flow, and structural stability. It consists of a dual-diameter geometry joined by a flat central disc and includes a dedicated internal air channel for pneumatic control.

Design Details:

- The connector was built from two cylindrical segments:
 - The rear cylinder has a diameter of 52 mm and a length of 11.5 mm, designed to match the end face of the linear actuator, sealed fit.
 - The front cylinder has a diameter of 45.79 mm, also 11.5 mm in length, intended to fit tightly into the radial shell's internal diameter (≈ 48 mm). This ensures a secure mechanical lock with minimal tolerance.
- Between these two cylinders is a flat disc section 4 mm thick, centered within this disc is a 20 mm diameter circular section, acting as a routing base for the internal air path, Inside this area, a 5 mm diameter through-hole runs longitudinally across the entire connector, connecting the actuator and balloon compartments. This unified air channel allows pressure from a single source to be distributed across both modules.

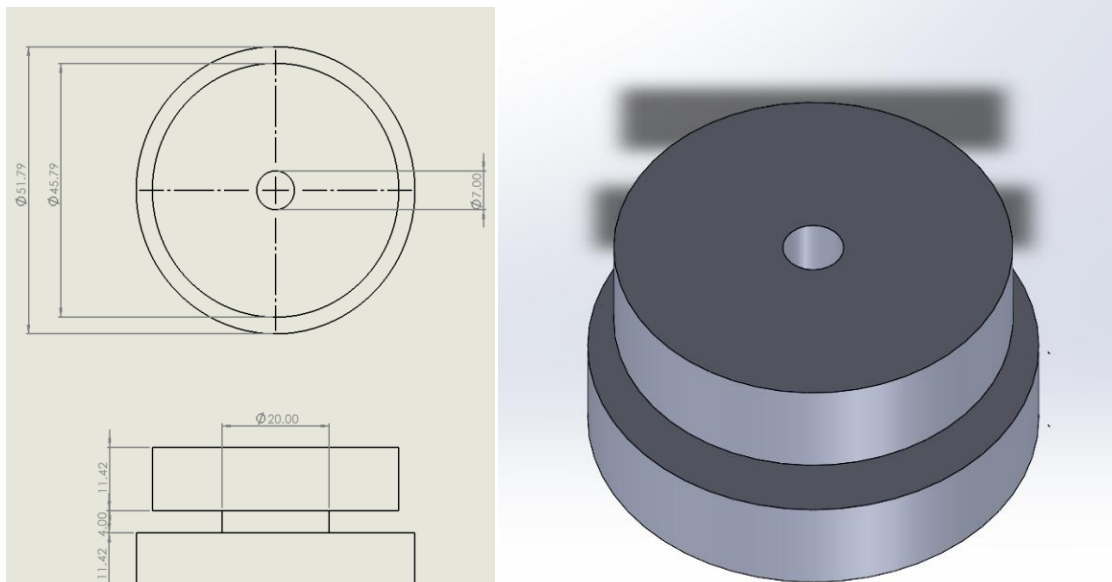


Figure 3.4.5 Connector Design

3.4.6 Internal Balloon Modelling

The balloon is the soft, inflatable anchoring component placed inside the radial shell. When pressurized, it expands through the grooves of the rigid shell to grip the pipe's internal surface, providing traction during crawling.

Design Details:

- The balloon was modeled as a cylindrical hollow structure in SolidWorks, designed to fit tightly within the radial shell's inner diameter (~48 mm) for accurate directional expansion.
- The outer diameter of the balloon was set to 48 mm, matching the shell's internal diameter. This tight fit ensures the balloon expands only outward through the grooves upon inflation, and not along unwanted paths.
- The wall thickness was kept at 4 mm, providing a balance between flexibility and pressure resistance, while also maintaining repeatability over multiple actuation cycles.
- The overall extrusion height was set to 42 mm, slightly shorter than the shell height (50 mm), allowing clearance for air hose routing and mechanical seating.
- Both ends of the balloon were sealed using solid faces to contain the internal pressure fully.
- A 5 mm diameter central through-hole was created along the axial direction to serve as the air inlet. This hole is aligned with the connector's pneumatic channel, allowing air to flow directly from the control unit into the balloon chamber.

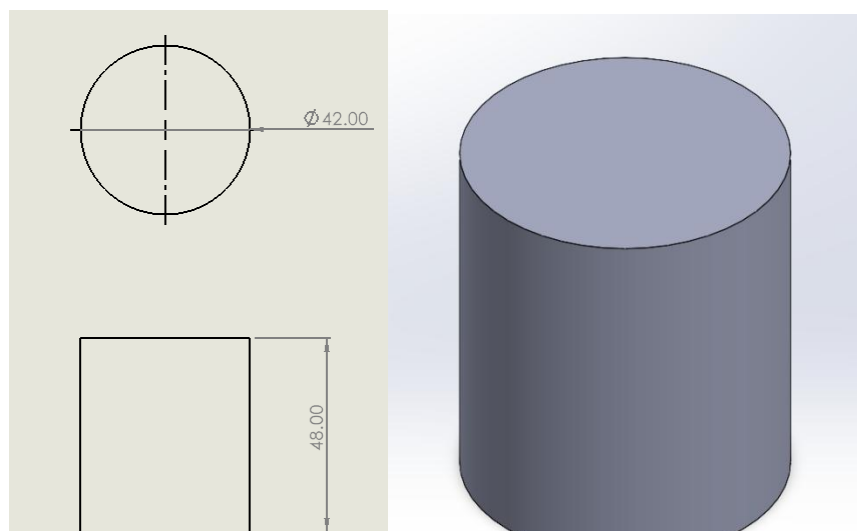


Figure 3.4.6 Balloon Design

3.5 Final Device Assembly Overview

Assembly Components and Configuration

Table 3.5 Final Device Component List

Item No.	Component	Quantity	Function
1	Linear actuator	1	Provides axial motion by elongating when inflated
2	Connector	2	Interfaces between linear actuator and radial shells, and to ensure structural attachment
3	Radial shell	2	Houses the balloon while it expands outward to grip pipe walls
4	Balloon	2	Inflates within the radial shells to provide anchoring during motion

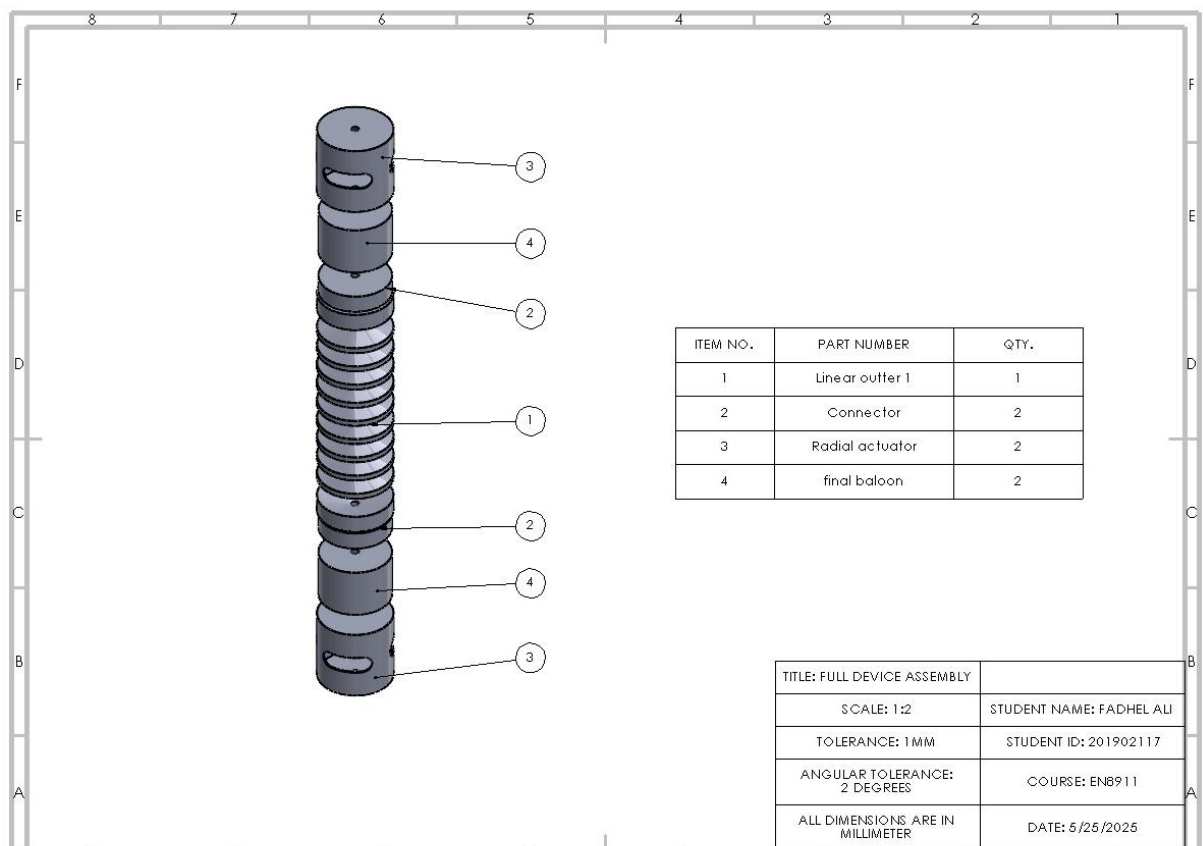


Figure 3.5 Final Device Assembly

3.6 Mold design

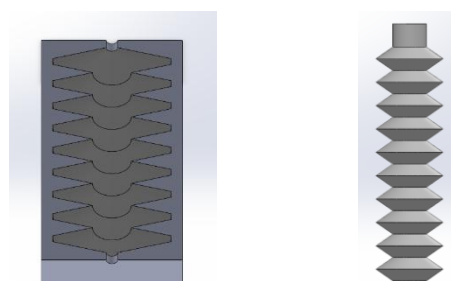
3.6.1 Linear Actuator Mold Design Iteration

The mold for the linear actuator was designed as a core and cavity system, using SolidWorks to replicate the final dimensions of the actuator. The goal was to fabricate the soft silicone structure with precise geometry while allowing consistent demolding and reusability.

Initial Mold Concept

The first mold design began as a rectangular block split into two halves (top and bottom). The workflow proceeded as follows:

1. A base mold block was created for both halves using *Extruded Boss/Base* with appropriate outer dimensions.
2. The finalized actuator model was positioned centrally between both mold halves.
3. The *Insert > Mold > Cavity* feature was used to subtract the actuator shape from the mold blocks.
4. To ensure added wall thickness and ease of demolding, the cavity was scaled up by 20% in the X, Y, and Z directions within the cavity feature settings.
5. A core insert was created and positioned into the internal bellow profile to prevent collapse and control internal hollowing.
6. The final mold was set up as a core-and-cavity type, with:
 - Cavity half housing the expanded outer surface of the actuator
 - Core half managing the internal volume and shape of the acutator



*Figure 3.6.1 Linear Actuator
Initial Mold Design*

Challenges Observed in First Trial:

- Wall Thickness Overcompensation: The 20% expansion will lead to dimensional differences
- Tight Fit at Core Ends: The 2 mm reduction in core dimension was sometimes insufficient, resulting in difficult demolding.
- Inconsistent Alignment: The rectangular shape provided no self-alignment geometry, leading to slight offsets between top and bottom halves during clamping.

3.6.2 Linear Actuator Mold Design Iteration

Following the limitations observed during the initial core-and-cavity design trial, a more effective mold strategy was adopted for the linear actuator. Instead of generating a separate core part, the improved approach utilized the final actuator geometry directly, split across the midplane, and embedded into a standard rectangular mold block.

Design Process:

- The finalized linear actuator (55 mm diameter, 130 mm length) was cut along the longitudinal axis, resulting in a symmetrical half. Later, the half was subtracted from a rectangular mold block using the *Cavity* feature to form a negative cavity matching half of the actuator. To achieve a 2 mm uniform wall thickness, the mold cavity was offset inward by 2 mm from the final actuator dimensions before subtraction.
- The outer mold remained rectangular for easy printing and handling.

Alignment Features:

To address misalignment issues seen in the first trial:

- The lower mold block was designed with cylindrical female holes, while the upper mold contained matching male alignment pins. These features ensured precise reassembly of both halves and prevented movement during casting.
- The final mold design was 3D printed in PLA.

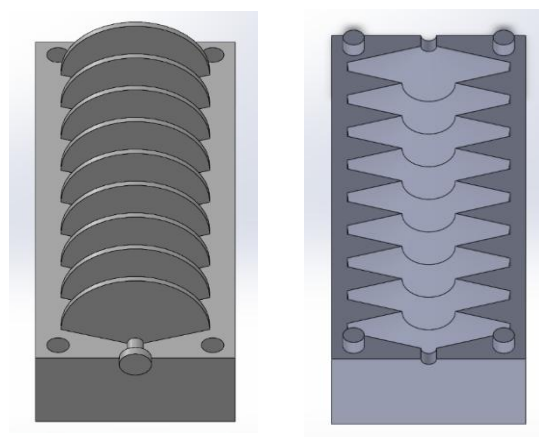


Figure 3.6.2 Linear Actuator Mold Design Iteration

3.6.3 Radial Balloon Mold Design

The balloon component, used for radial expansion and gripping. The balloon was cast using a core-and-cavity mold, designed to ensure 4 mm wall thickness and a sealed end.

Design Process:

- The outer mold cavity was created by extruding a circle with a 40 mm internal diameter to a height of 46 mm. To ensure safe demolding without damaging the balloon, the entire mold geometry was split into two equal halves, except the base.
- The core, centered within the cavity, was modeled with a 36 mm diameter and only 42 mm in height, making it 4 mm shorter than the cavity.
 - This intentional offset created a 4 mm-thick base cap at the top of the balloon once the silicone cured. As a result, one end of the balloon was automatically sealed during the casting process without requiring post-curing closure or bonding. The design ensured a uniform radial gap around the core achieving 4 mm wall thickness, while the top sealing layer was cast in one pour.
- Air Inlet Channel, 5mm hole was added through the center of the core to form the air inlet port, allowing for pneumatic inflation after curing. This hole aligned with the connector's central port, ensuring continuous airflow during robot operation.

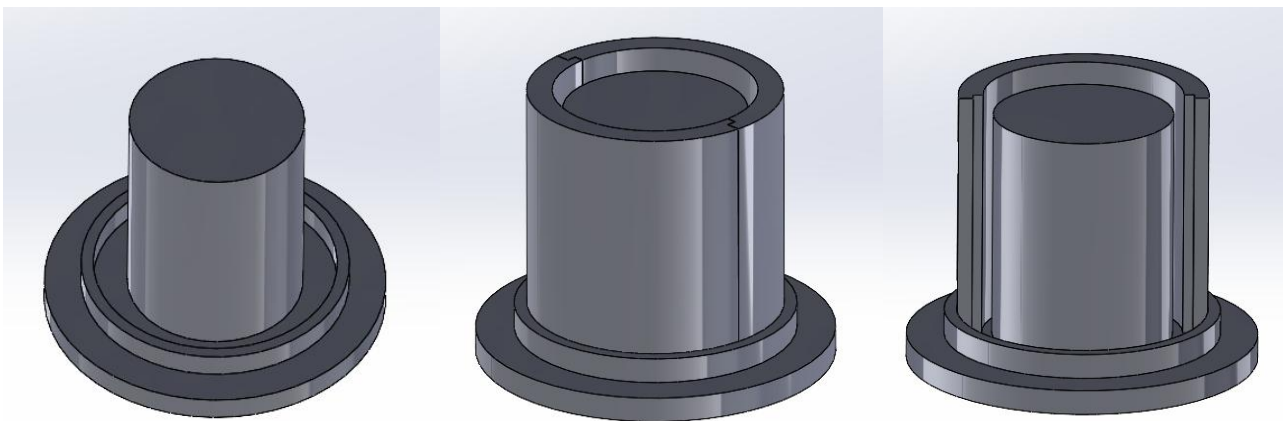


Figure 3.6.3 Balloon Mold

3.6.4 Balloon seal mold

Design Purpose:

- Provide a thicker, structurally reinforced end to reduce tear risk around the air inlet.
- Allow a dip-sealing technique, where the open end of the cured balloon is fully dipped in the silicone to ensure that both ends are completely sealed to avoid air leaks that can interfere with its function while in motion.
- The cured balloon's open end will be dipped into the cavity, which will be pre-filled with mixed silicone to form a 3–4 mm thick sealing layer.

Mold Geometry:

- The mold was modeled as a cylindrical cavity, with an internal diameter of 40 mm to match the outer diameter of the balloon.
- It was extruded to a height of 20 mm, forming a vertical chamber.

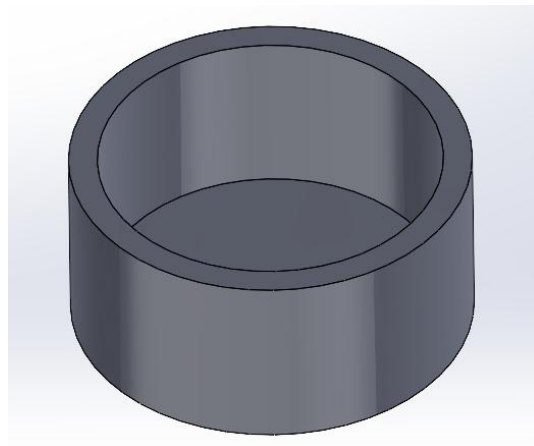


Figure 3.6.4 Balloon End Seal Mold

3.7 Fabrication of Molds using 3D Printers

To manufacture the mold sets used for casting the soft actuator components, Fused Deposition Modeling (FDM) 3D printing was employed.

3.7.1 Printer Setup

The molds were designed in SolidWorks 2023 and exported in .sldprt format. For printing preparation, the files were converted into .STL format and imported into Creality Slicer, the software used for the Creality Ender-3 V3 printer (0.4 mm nozzle).



Figure 3.7.1 Creality 3D Printer & Software

The software processed the 3D geometry and generated G-code instructions, which the printer used to construct the molds layer-by-layer.

Table 3.7.1 Printer Setup Parameters

Parameter	Setting
Layer height	0.2 mm (standard resolution)
Infill percentage	15–35% (based on part strength)
Support structures	Enabled
Filament type	Generic PLA (1.75 mm)

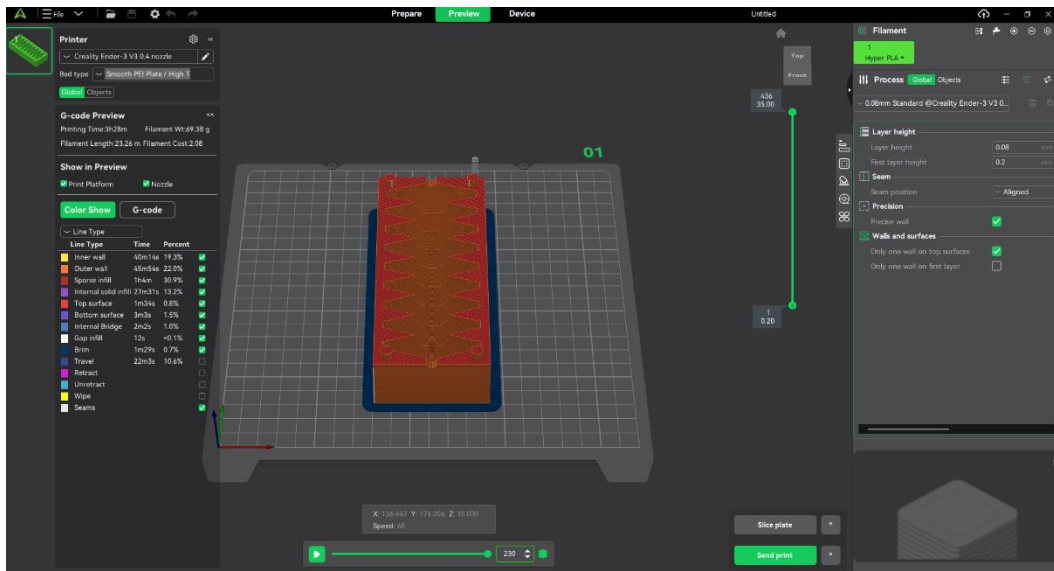


Figure 3.7.1 Linear Mold 3D Preparation

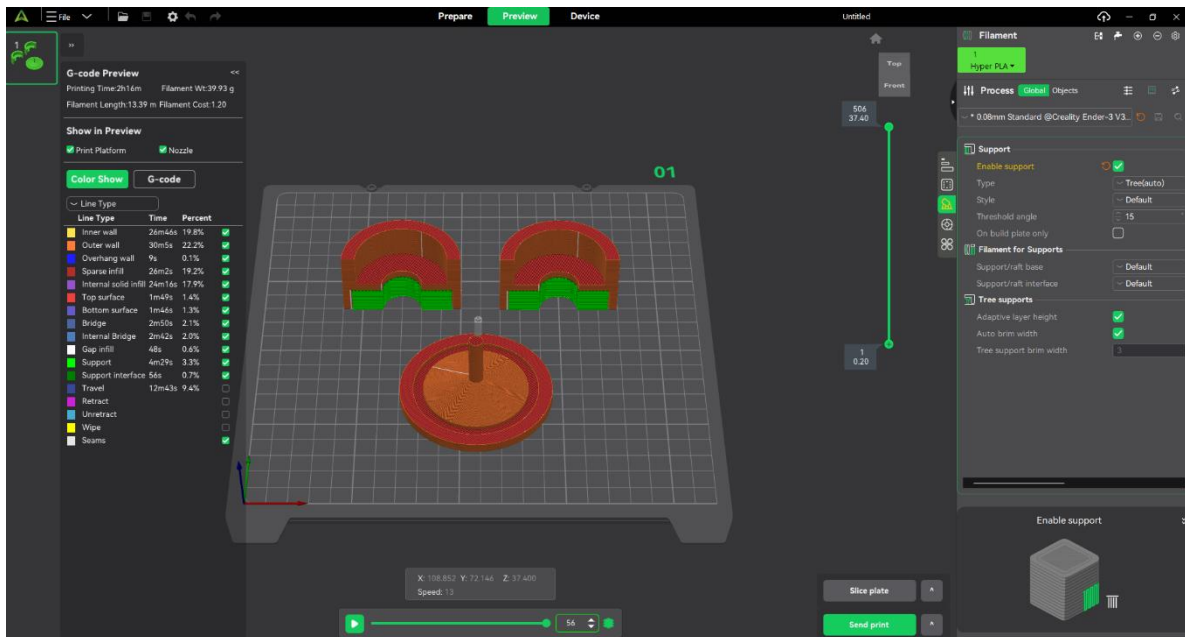


Figure 3.7.1 Balloon Mold 3D Preparation

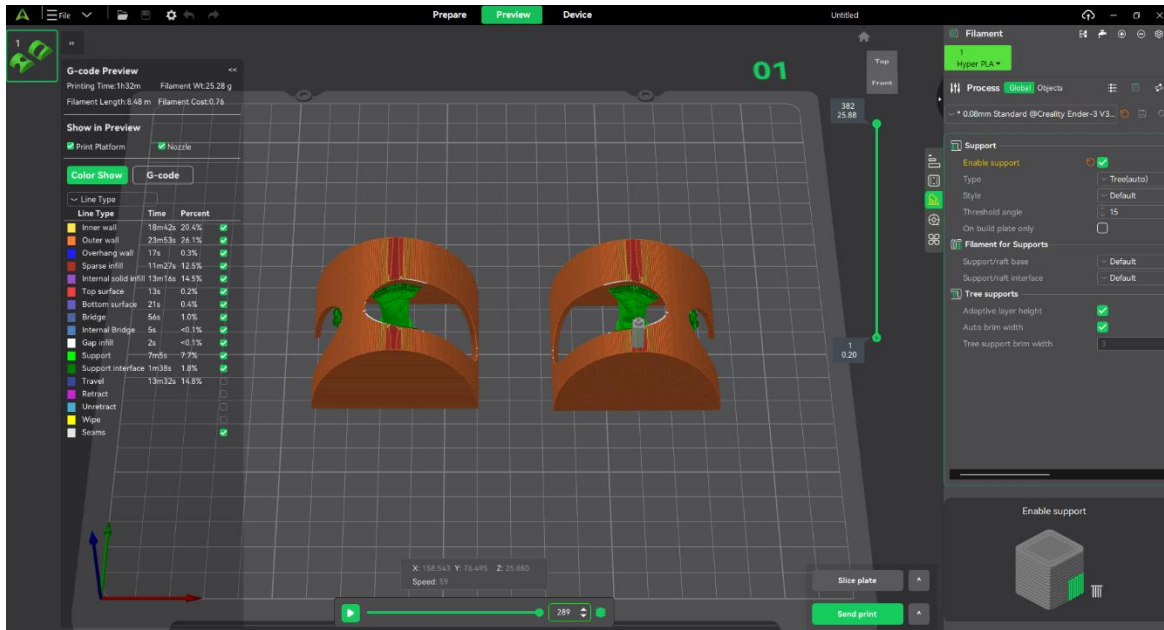


Figure 3.7.1 Radial Shell Mold 3D Preparation

- Supports (shown as green contour) were automatically added where needed to maintain overhanging features and internal geometries. Print times for each mold half ranged between 2–4 hours, depending on volume and complexity. Once slicing was complete, the file was transferred to the printer.

3.7.2 Fabrication Issues and Resolution

- The printed mold halves were slightly warped, leading to misalignment between top and bottom halves during silicone casting. This resulted in a final actuator with inconsistent wall thickness and poor bonding. The suspected cause was bed temperature fluctuation, which caused slight warping of PLA during printing.



Figure 3.7.2 Faulty Linear Mold Prints

To resolve this, the fabrication was repeated at Fab Lab Bahrain using their high-performance Bambu Lab X1 Carbon printer.



Figure 3.7.2 BambuLab X1 Carbon 3D Printer

The same STL files and Generic PLA filament were used, but the output quality was significantly improved. The new molds printed on the Bambu Lab machine showed perfect flatness, clean edge definition, and tight dimensional accuracy, eliminating the earlier alignment issues.

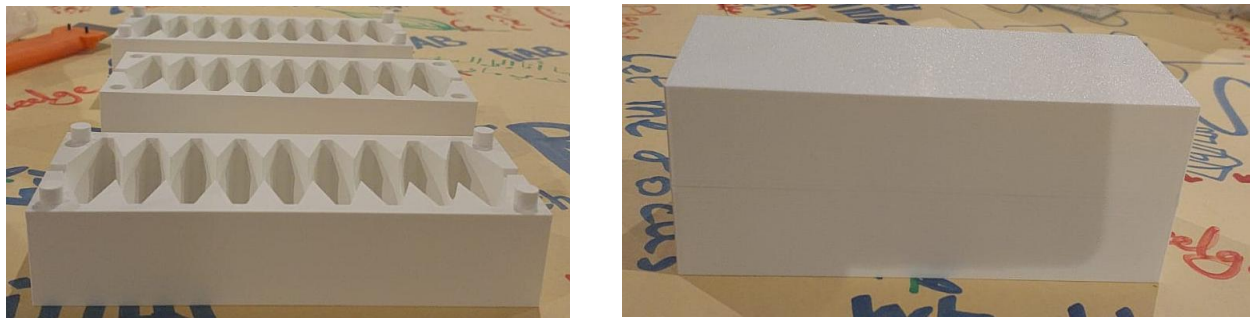


Figure 3.7.2 Ideal Mold Prints (Linear)

3.8 Simulation

3.8.1 Step 1: Material Assignment – Hyperelastic Yeoh Model for DoYB Silicone

The simulation began by creating a Static Structural study, which was selected due to its suitability for analyzing nonlinear deformations under steady-state pressure loading. While soft robotic actuation is naturally dynamic, the static structural solver remains an appropriate choice for evaluating maximum deformation, strain limits, and stress behavior.

For the material assignment, the focus was on replicating the real-world mechanical behavior of the DoYB silicone used in the linear actuator's structure. Since silicone exhibits large elastic strains, a hyperelastic material model was required to accurately simulate their nonlinear stress-strain relationship.

1. From the ANSYS Engineering Data library, the Hyperelastic Material category was selected.
2. The Yeoh 3rd Order Model was chosen, using the “Elastomer Sample (Yeoh)”
 - The Yeoh model is ideal for nearly incompressible, rubber-like materials and captures large-strain behavior effectively using a strain energy potential based solely on the first strain invariant. Compared to Mooney-Rivlin or Ogden models, the Yeoh formulation is simpler yet accurate, making it suitable for the deformation modes in this actuator.

The following material properties were assigned:

Table 3.8.1 Material Properties

Property	Value
Density	1049 kg/m ³
Hyperelastic Model Type	Yeoh – 3rd Order
Yeoh Constants (MPa)	C10 = 0.127, C20 = -0.149, C30 = 0.153 <i>(from supplier data sheet)</i>
Incompressibility Parameters	D1 = 0, D2 = 0, D3 = 0 <i>(assumed fully incompressible)</i>

Outline of Schematic A2: Engineering Data				
	A	B	C	D
1	Contents of Engineering Data		Source	Description
2	Material			
3	Elastomer Sample (Yeoh)		Hyp	Sample data to model as Yeoh (courtesy of Axel Products, Inc. http://www.axelproducts.com)
*	Click here to add a new material			

Properties of Outline Row 3: Elastomer Sample (Yeoh)				
	A	B	C	D
1	Property	Value	Unit	E
3	Density	1049	kg m ⁻³	
4	Uniaxial Test Data	Tabular		
5	Scale	1		
6	Offset	0	Pa	
7	Biaxial Test Data	Tabular		
8	Has Lateral Strain	No		
9	Scale	1		
10	Offset	0	Pa	
11	Shear Test Data	Tabular		
12	Has Lateral Strain	No		
13	Scale	1		
14	Offset	0	Pa	
15	Yeoh 3rd Order			
16	Material Constant C10	0.127	MPa	
17	Material Constant C20	-0.149	MPa	
18	Material Constant C30	0.153	MPa	
19	Incompressibility Parameter D1	0	Pa ⁻¹	
20	Incompressibility Parameter D2	0	Pa ⁻¹	
21	Incompressibility Parameter D3	0	Pa ⁻¹	

Figure 3.8.1 DoYB Material Properties (ANSYS)

3.8.2 Step 2: Geometry Import and Internal Face Selection

After defining the material model, the next phase involved preparing the linear actuator geometry for simulation. The part was initially modeled in SolidWorks 2023 and exported as a (.STEP) file format to ensure compatibility with ANSYS.

- The geometry was imported into ANSYS DesignModeler, where it preserved all external and internal contours of the bellows actuator. Due to the actuator's hollow internal chamber, it was essential to isolate the internal pressure-bearing surfaces.
1. To achieve this, the Section Plane tool was used to slice the actuator model along its central longitudinal axis, exposing the internal cavity where air pressure would act during operation.
 2. Once sectioned, the Named Selection tool was employed to manually select all interior surfaces that form the air chamber walls:
- These selected faces included the inner curves of the bellows, the rear face near the inlet, and the inner wall of the actuator's front cap. Assigning these surfaces as a single named selection ("innerwalls"), ensuring pressure was only applied internally, avoiding incorrect load on the external structure.

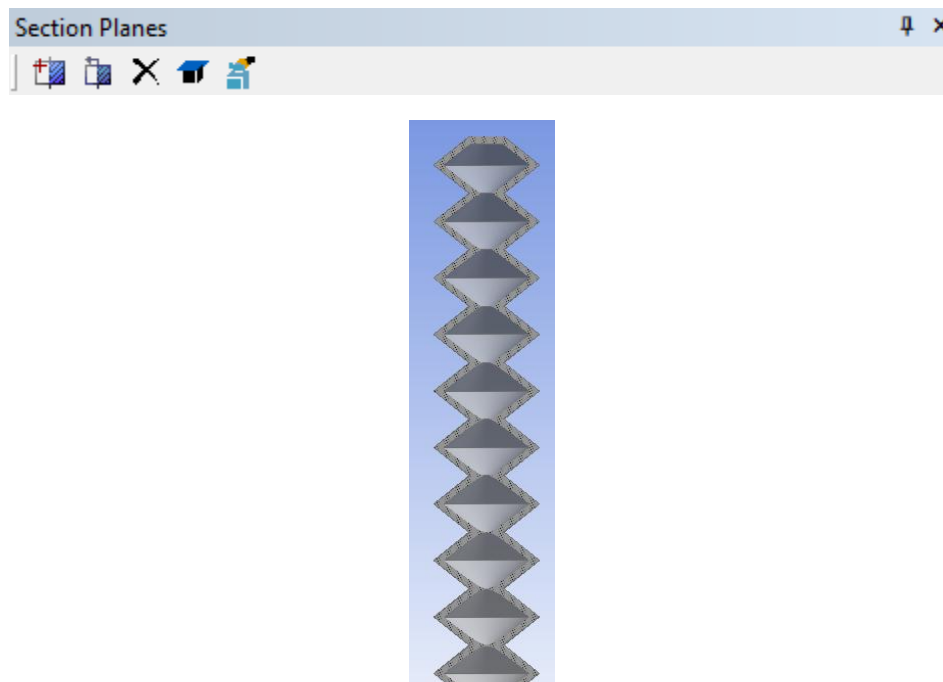
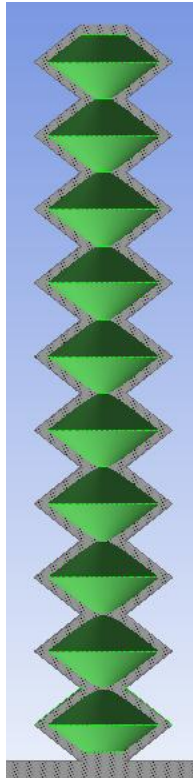


Figure 3.8.2 Section View (Linear)

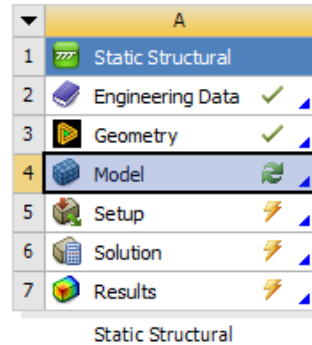


Sketching		Modeling
Details View		4
☐	Details of NamedSel2	
	Named Selection	innerwalls
	Geometry	Not selected
	Propagate Selection	Yes
	Export Selection	Yes
	Include In Legend	Yes

Figure 3.8.2 Named Selection (Linear)

3.8.3 Step 3: Assigning Material in Mechanical Model

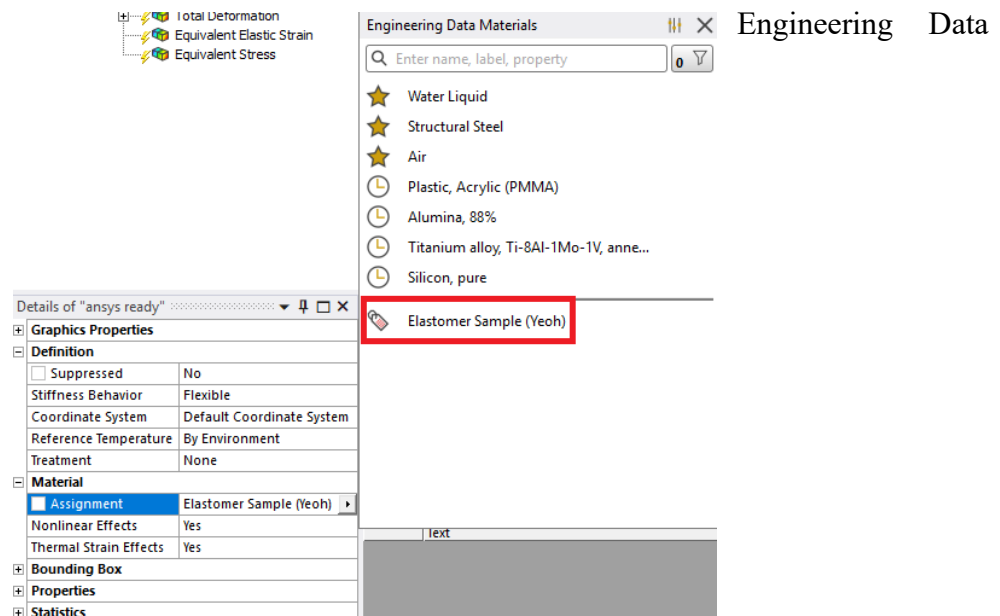
After defining the geometry and setting up the "innerwalls" named selection in DesignModeler, the model was transitioned to the ANSYS Mechanical interface for simulation preparation.



Upon entry into the Model workspace, the next key task was to assign the previously defined hyperelastic material to the imported geometry.

- The imported actuator part appeared as a single body under the Geometry tree. And, the material created in Step 1 (DoYB – Hyperelastic (Yeoh)) was now available under the

library.



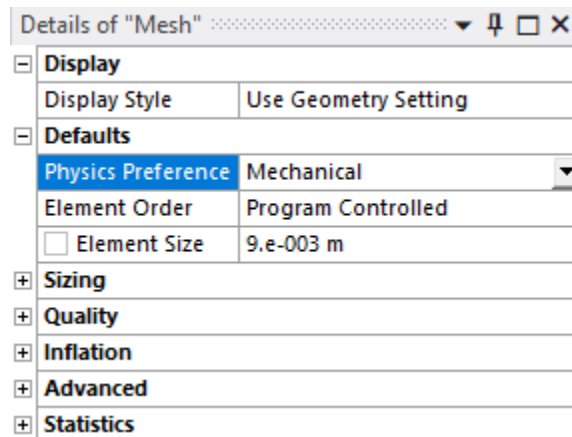
- This material was then assigned to the actuator body via the “Material Assignment” section in the Details panel.

3.8.4 Step 4: Mesh Settings and Element Control

With the material assigned, the next phase in ANSYS Mechanical involved generating a finite element mesh that could accurately resolve large, nonlinear deformations while remaining computationally efficient.

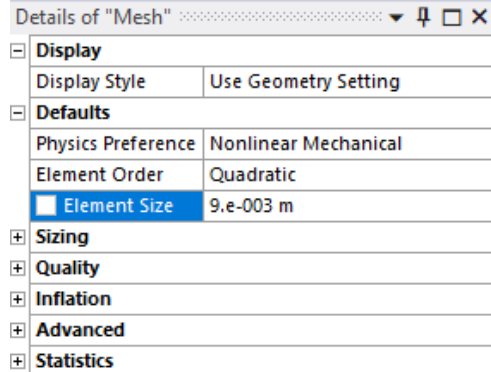
Mesh Control Parameters:

- The default Physics Preference was initially set to “Mechanical”, which is suited for standard linear material behavior and small-strain problems.



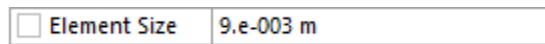
- However, to better represent the behavior of hyperelastic silicone under high strain, this was manually updated to “Nonlinear Mechanical”. This change optimizes meshing algorithms and element formulations for handling large-strain, nonlinear material behavior such as rubber or elastomers.

- The Element Order was also changed from “Program Controlled” to “Quadratic”:
 - Quadratic (second-order), allows better curvature approximation of bellow surfaces. They also improve accuracy in simulations involving nonlinear geometry and material models, which was essential for capturing the bending and expansion of the actuator’s walls during inflation.



Element Size and Refinement:

The global element size was manually set to 0.009 m (9 mm).



- This size was chosen as a balance between mesh resolution and the system’s efficiency. It was fine enough to capture local curvature in the bellows, but not so fine that it would cause excessive solver time or overload the CPU memory. If the mesh were any finer, particularly below 0.005 m, the system could become unstable, especially during nonlinear substeps or contact-based iterations, leading to long runtimes or simulation crashes.

3.8.5 Step 5: Static Structural Settings

With meshing completed, the next step involved configuring the solution environment in ANSYS Mechanical to accurately reflect the physical behavior of the soft actuator during inflation. This included enabling nonlinear deformation handling and applying realistic environmental loads.

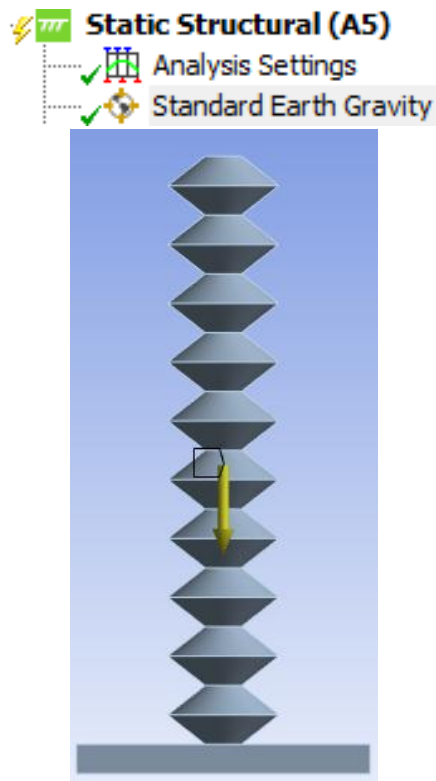
Enabling Large Deflection

- In the Analysis Settings, the option “Large Deflection” was enabled. This is essential when the structure is expected to undergo geometric nonlinearity, such as Large elastic stretching, Nonlinear shape changes and Significant displacement relative to the original dimensions. Without this setting, the solver assumes a linear geometry, meaning any large deformation would be inaccurately represented or lead to convergence failure. In soft robotics, large deflections are typical due to the high elasticity of materials like DoYB, making this setting critical for simulation reliability.

☐ Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	On

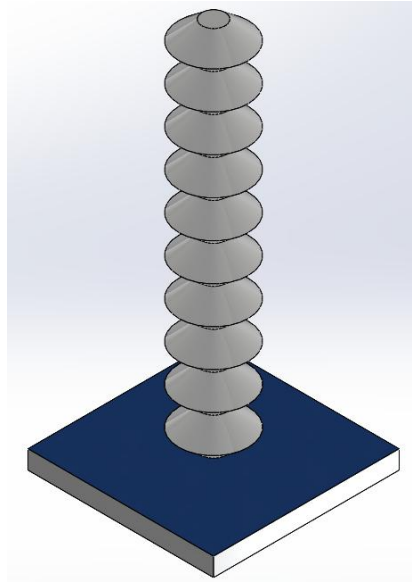
Applying Gravity

- A gravity load was added in the $-Y$ direction, as the actuator was modeled in a vertical orientation to simulate its real-world positioning during inflation. Applying gravity is important for capturing the natural downward pull on the actuator body and the effect of self-weight on deformation (especially relevant for silicone-based materials which can flex under their own mass)
- The standard gravity value used was 9.81 m/s^2 , applied uniformly across the entire body.

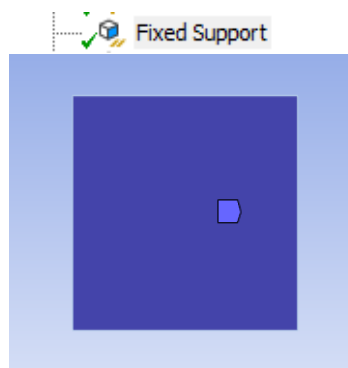


Applying Fixed Support at the Base

A non-physical base feature was added in the initial SolidWorks sketch to serve as an anchoring surface for the simulation.



- This flat square base was not part of the real design and is excluded from manufacturing but was necessary for simulation accuracy. Without this base, applying a fixed support directly to the actuator's end face would have resulted in:
 - Nonphysical stress concentrations
 - Contact interference
 - Solver convergence errors and high local distortion
- A Fixed Support was applied to the opposite end face of this artificial base, effectively holding the actuator in place and isolating deformation to the flexible body, especially the bellows section.



Pressure Boundary Condition

- A pressure boundary condition was applied to the internal surface of the actuator via the named selection "innerwalls" created in DesignModeler.



Details of "Pressure"	
= Scope	
Scoping Method	Named Selection
Named Selection	Innerwalls

- The pressure value was set to 100,000 Pascals (100 kPa), which reflects a moderate pneumatic load typically used in soft robotic inflation tests.

= Definition	
Type	Pressure
Define By	Normal To
Applied By	Surface Effect
Loaded Area	Initial
☒ Magnitude	1.e+005 Pa (ramped)
Suppressed	No

- This pressure was applied uniformly to all internal faces of the actuator body, replicating the condition of air being pumped into the internal chamber. This setup enabled accurate analysis of axial expansion, stress localization and strain distribution during actuation.

3.8.6 Step 6: Solution Setup and Result Requests

Once the boundary conditions were defined, the Solution section in ANSYS Mechanical was configured to extract important performance indicators from the simulation. These outputs were chosen specifically to evaluate how the soft linear actuator behaves under internal pressure and self-weight, with a focus on deformation, material strain, and stress.

1. Total Deformation

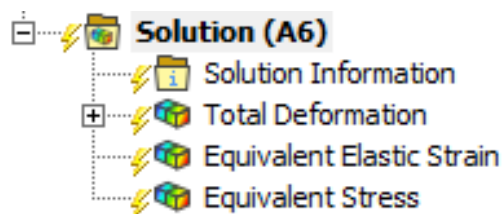
- This result type measures the overall displacement magnitude of the model. It was used to evaluate the elongation of the actuator and its deformation. Key for validating that the actuator can safely stretch within expected limits, especially for comparing simulation to physical prototype results.

2. Equivalent Elastic Strain (Von-Mises)

- The Equivalent Elastic Strain result calculates strain energy density using Von-Mises criteria and is ideal for hyperelastic soft materials. It indicates how much elastic deformation the silicone undergoes critically. This helps identify localized overstretch zones, which are common in soft robotic utilizing the same material and can lead to material fatigue or rupture.

3. Equivalent Stress (Von-Mises)

- The Von-Mises Equivalent Stress was added to assess the overall stress distribution within the actuator. Even though silicone does not fail like rigid materials, this metric is useful for comparing silicone's tensile strength limit, Identifying stress concentration zones and evaluating if the applied 100 kPa pressure remains within safe operating ranges.



3.8.7 Step 7: Final Mesh Generation and Solution

With all boundary conditions, material model, and result tools configured, the final step involved generating the mesh and launching the solver to compute the simulation results.



3.9 Simulation of Balloon Expansion in ANSYS

To evaluate the radial expansion behavior of the internal balloon component, the same simulation workflow used for the linear actuator was repeated with appropriate geometric and material adjustments. The goal was to assess the balloon's outward deformation, strain behavior, and stress zones when subjected to pneumatic pressure while enclosed inside the radial shell.

3.9.1 Static Structural Study Creation

- A new Static Structural analysis system was created in ANSYS 2023 R1.

3.9.2 Material Assignment – ECOFLEX00-10 Silicone

- The same hyperelastic Yeoh 3rd order model (defined earlier) was assigned to the balloon body.

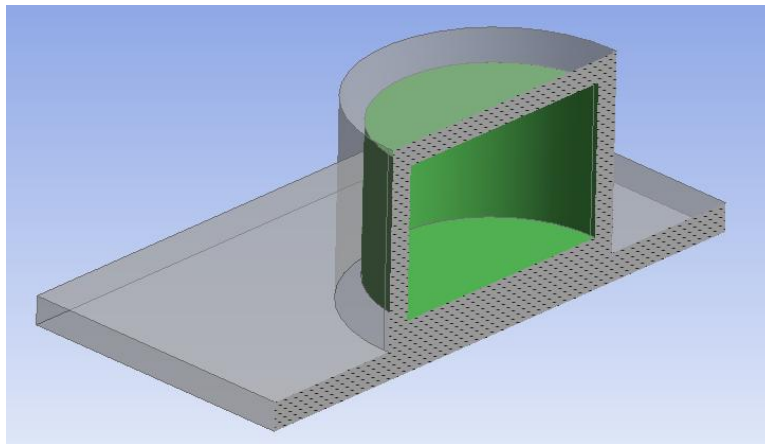
Table 3.9.2 Material Properties (ECOFLEX00-10)

Property	Value
Density	1049 kg/m ³
Hyperelastic Model Type	Yeoh – 3rd Order
Yeoh Constants (MPa)	$C10 = 1.00 \times 10^{-3}$, $C20 = 1.31 \times 10^{-2}$, $C30 = 2.056 \times 10^{-5}$ (from supplier data sheet)
Incompressibility Parameters	$D1 = 0$, $D2 = 0$, $D3 = 0$ (assumed fully incompressible)

Properties of Outline Row 3: Elastomer Sample (Yeoh)				
	A	B	C	D E
1	Property	Value	Unit	
3	Density	1049	kg m ⁻³	
4	Uniaxial Test Data	Tabular		
5	Scale	1		
6	Offset	0	Pa	
7	Biaxial Test Data	Tabular		
8	Has Lateral Strain	No		
9	Scale	1		
10	Offset	0	Pa	
11	Shear Test Data	Tabular		
12	Has Lateral Strain	No		
13	Scale	1		
14	Offset	0	Pa	
15	Yeoh 3rd Order			
16	Material Constant C10	0.001	MPa	
17	Material Constant C20	0.0131	MPa	
18	Material Constant C30	2.056E-05	MPa	
19	Incompressibility Parameter D1	0	Pa ⁻¹	
20	Incompressibility Parameter D2	0	Pa ⁻¹	
21	Incompressibility Parameter D3	0	Pa ⁻¹	

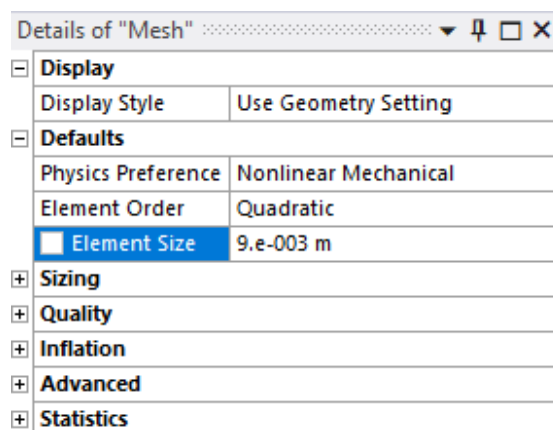
3.9.3 Geometry Import

- The balloon geometry was exported from SolidWorks as a STEP file and imported into ANSYS DesignModeler.
- A section plane was used to expose the balloon's hollow interior.
- Using the Named Selection tool, all internal faces of the balloon's chamber were selected and grouped under "innerwalls" to serve as the pressure application zone.



3.9.4 Mesh Generation and Element Setup

- The Physics Preference was set to Nonlinear Mechanical, and Element Order was switched to Quadratic to handle curvature.
- The element size was again set to 0.009 m, balancing solver accuracy with computer performance.

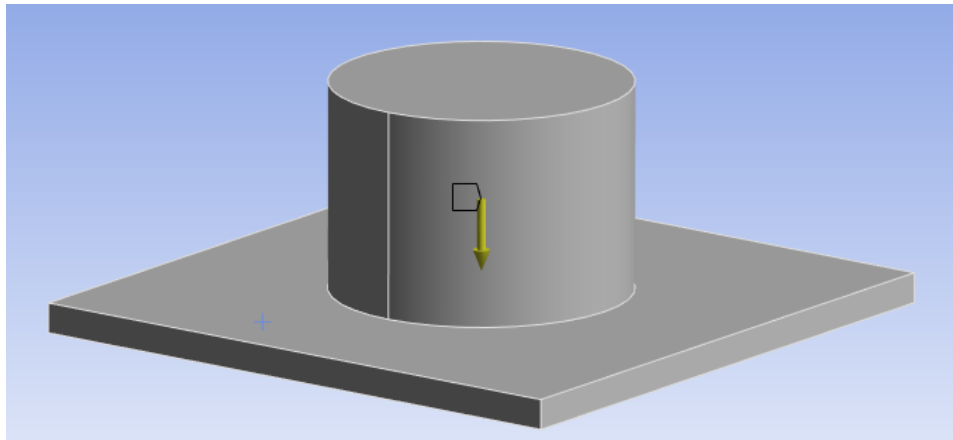


3.9.5 Boundary Conditions

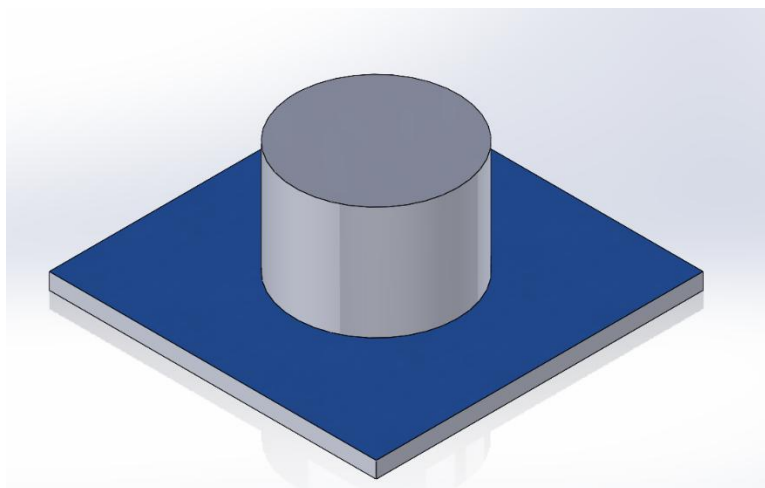
- Large Deflection was enabled to account for significant radial deformation.

Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	On

- Gravity was applied in the $-Y$ direction.



- A Fixed Support was applied at the rear face of a small simulated base used only for anchoring the balloon during inflation. Similar to the linear part, the base was not part of the real balloon design but was added to prevent boundary distortion errors.



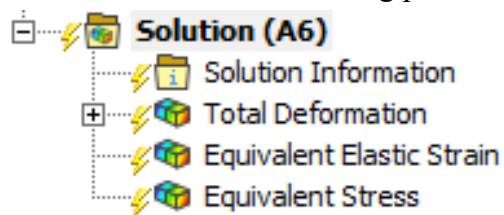
3.9.6 Pressure Loading

- A uniform internal pressure of 100,000 Pa (100 kPa) was applied to the "innerwalls" named selection.

Definition	
Type	Pressure
Define By	Normal To
Applied By	Surface Effect
Loaded Area	Initial
☐ Magnitude	1.e+005 Pa (ramped)
Suppressed	No

3.9.7 Run simulation

- The solver has been executed to obtain the following parameters:

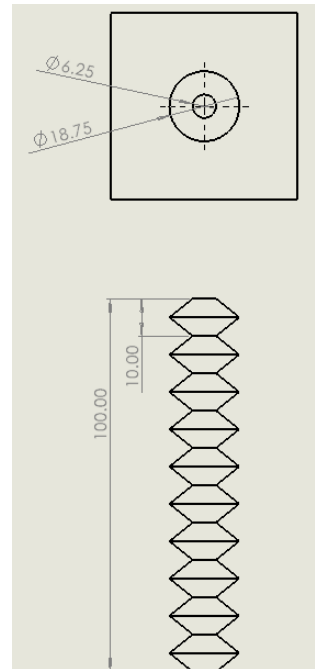


4 Results

4.1 Early Design Limitations (Linear)

An initial linear actuator design consisting of 10 closely packed bellows ($\text{Ø}18.75\text{ mm}$) with thin 1 mm walls and a narrow 2 mm pressure inlet was modeled in ANSYS. However, repeated simulation attempts failed to converge, even at low pressures (1–100 kPa), due to extreme non-linear behavior, element distortion, and solver instability. The error logs consistently identified mesh collapse and excessive deformation during early substeps. Despite correct boundary conditions, the geometry's thin walls and high bellow count contributed to poor aspect ratios and mesh distortion under hyperelastic strain, rendering the design unviable for continued simulation or fabrication.

Parameter	Value
Total Length	100 mm
Bellows Count	10
Bellow Length (each)	10 mm
Bellow Diameter	18.75 mm
Inlet Hole Diameter	6.25 mm
Wall Thickness	1 mm



Despite following identical setup steps (material assignment, boundary conditions, mesh, pressure application), the simulation for this geometry consistently failed with the following issues:

Messages		
☒ Show Errors (2) ☒ Show Warnings (3) ☒ Show Info (0) ☒ Merge Messages ☒ Pop-up Messages		
	Text	Association
Warning	Although the solution failed to solve completely at all time points, partial results at some	Project>Model>Static Structural>Solution
Error	The solver engine was unable to converge on a solution for the nonlinear problem as cor	Project>Model>Static Structural>Solution
Warning	The solution failed to solve completely at all time points. Restart points are available to cc	Project>Model>Static Structural>Solution
Warning	The solution failed to solve completely at all time points. Restart points are available to continue the analysis.	Project>Model>Static Structural>Solution
Error	Element 746 located in Body 'ansys ready' (and maybe other elements) has become highly	Project>Model>Static Structural>Solution

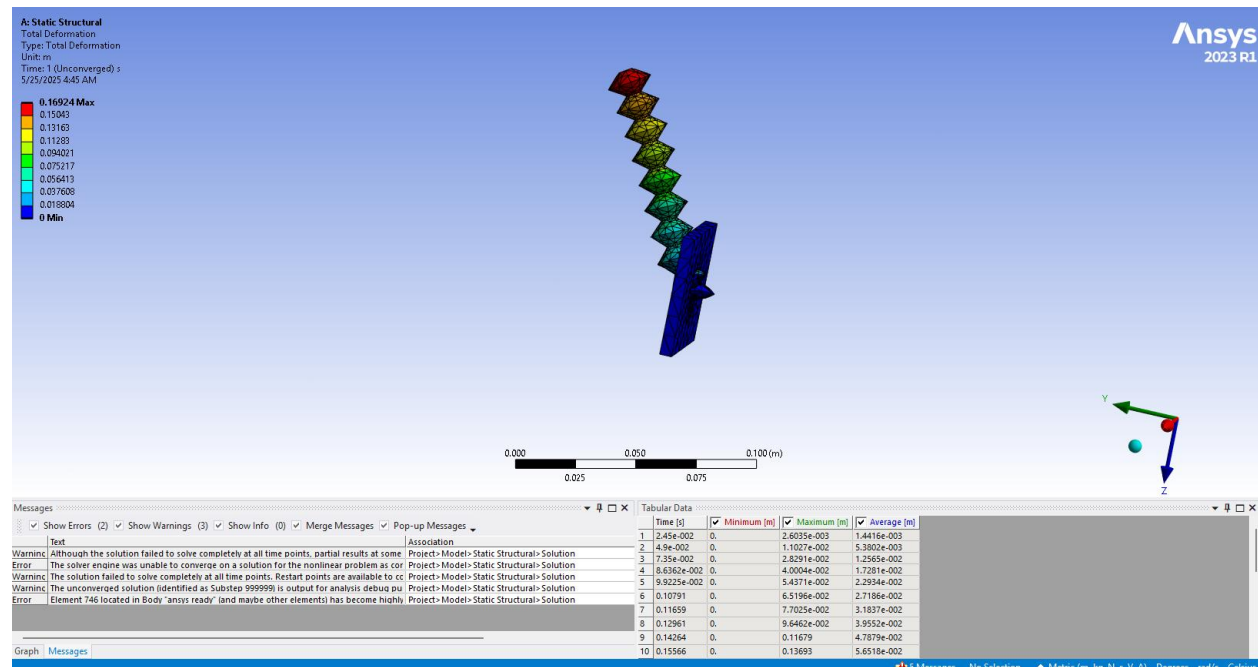
- ANSYS

Error:

"Element 746 located in Body 'ansys ready' has become highly distorted."
- Solver

Output:

"The unconverged solution (Substep 999999) is output for debug only."
- Stress/strain behavior: Solver cannot converge due to over-deformation or unstable mesh



4.2 Finalized Linear Actuator design/model

The final design of the linear actuator underwent minor but essential modifications to improve simulation stability and structural response under pressure. With nine bellows and adjusted geometry, the component was modeled and analyzed using ANSYS Static Structural to evaluate its deformation, material's strain and stress performance. This simulation aims to assess how the actuator responds to pneumatic expansion using a hyperelastic material model and to verify whether the design delivers effective elongation without compromising material integrity. More detailed performance insights and comparisons with earlier prototypes are discussed in the following sections.

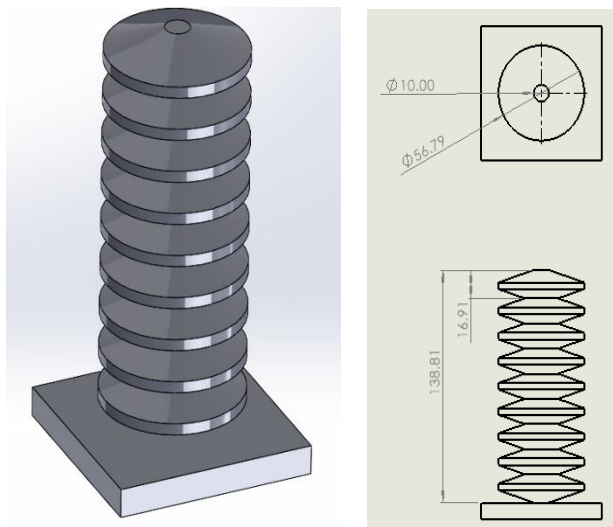


Table 4.2 Final Linear Design Parameters

Parameter	Value
Total Length	138.81 mm
Bellows Count	9
Bellow Length (each)	16.91 mm
Bellow Diameter	56.79 mm
Inlet Hole Diameter	10 mm
Wall Thickness	2.5 mm

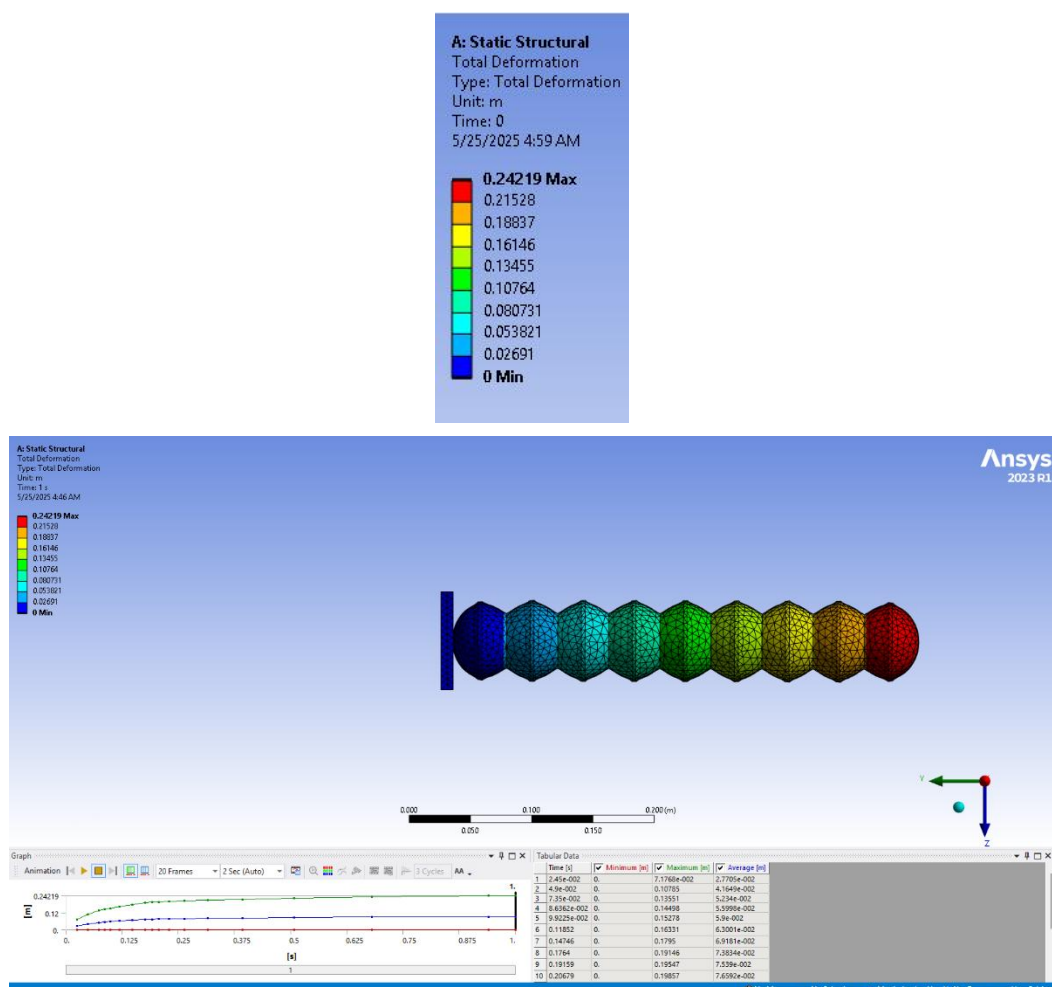
4.2.1 Deformation calculation

Total deformation analysis was conducted to evaluate the actuator's elongation behavior under internal pressure. This study was crucial because the soft crawler's locomotion depends on controlled, repeatable axial expansion. By measuring displacement across the structure, the simulation confirmed whether the actuator could achieve the required stroke length and maintain structural integrity throughout its cycle. It also helped detect any signs of excessive bulging or uneven extension that might affect its movement efficiency.

Obtained results from ANYS & SolidWorks:

Original (undeformed) length of actuator $L_0 = 138.9 \text{ mm}$

Maximum deformation (from ANSYS) $\Delta L = 0.24219 \text{ m} = 242.19 \text{ mm}$



Final Length

$$L_{Final} = L_0 + \Delta L$$

$$138.9 + 242.19 = 381.09 \text{ mm}$$

Elongation Percentage

$$L_{Final} = \frac{\Delta L}{L_0} \times 100$$

$$\frac{242.19}{138.8} \times 100 = 174.36\%$$

Justification for Total Deformation Results

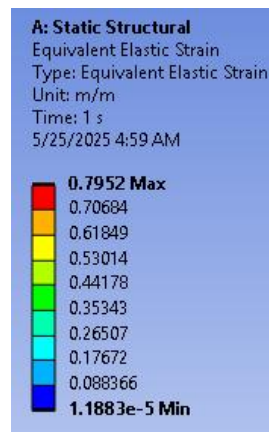
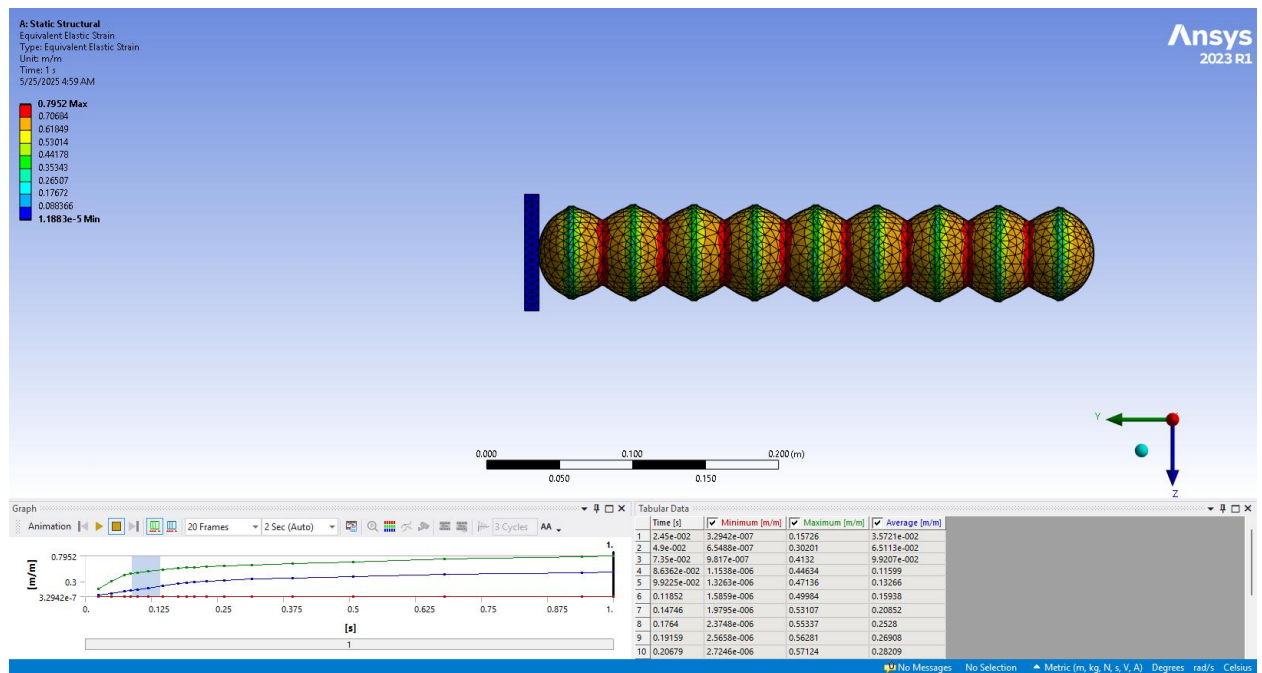
The total deformation simulation of the final linear actuator, modeled in ANSYS using a Yeoh 3rd-order hyperelastic material model, accurately demonstrates the actuator's ability to stretch under internal pneumatic pressure. The observed maximum deformation of 242.19 mm, when added to the original length of 138.9 mm, results in a final extended length of 381.09 mm equating to a 174.36% elongation, nearly three times its original length.

This outcome is both expected and justified for several reasons:

- The actuator is made from Ecoflex™ 00-10, a soft silicone rubber with a known elongation capacity of 800% before rupture (Smooth-On, 2023).
- Fixed support at the base correctly simulates anchoring one end of the actuator (like being attached to the radial shell while the balloon is actuated).
- The deformation is stable (no folding or mesh error), indicating strength under load.

4.2.2 Material strain data

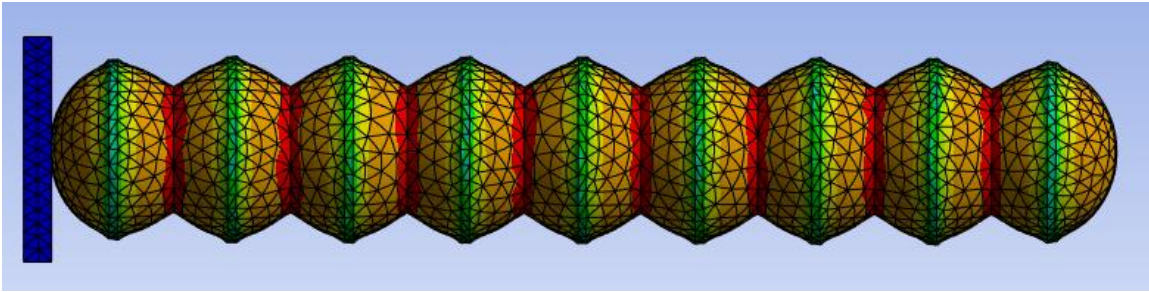
Elastic strain simulation was essential to assess how much the material stretches during actuation and whether it remains within its elastic recovery range. For materials like Ecoflex 00-10 and DoYB silicone, it was critical to ensure that strains did not exceed their safe limits to avoid permanent deformation or fatigue. This study identified localized high-strain regions, providing feedback for optimizing geometry and confirming the actuator's suitability for repeated use.



Results summary

Table 4.2.2 Material Strain Result Summary

Measured	Value
Max Elastic Strain ϵ_{max}	0.7952 (m/m)
Avg Elastic Strain $\epsilon_{average}$	0.28209 (m/m)
Min Elastic Strain $\epsilon_{minimum}$	3.29×10^{-7} (m/m)

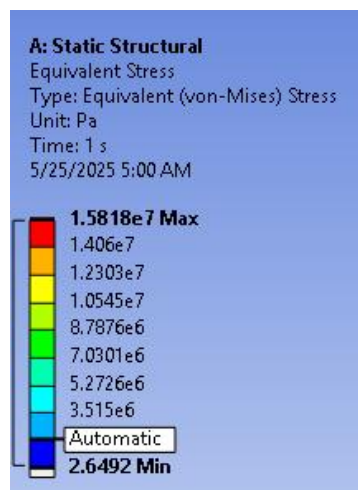
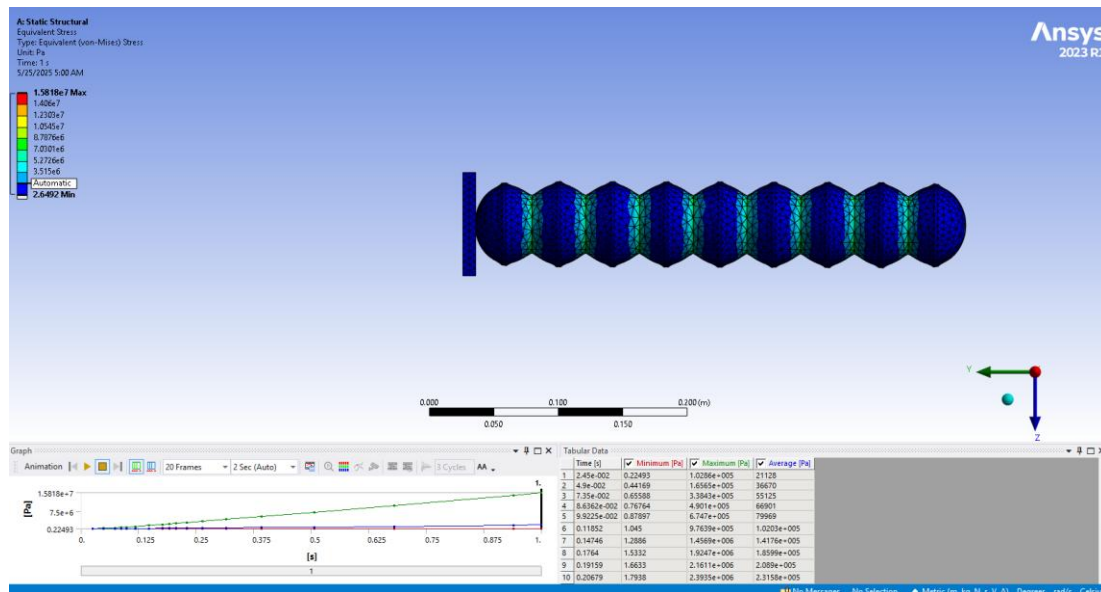


As illustrated in the figure above, the regions highlighted in red represent zones of maximum elastic strain, indicating the portions of the actuator most responsible for the overall elongation during pressurization. These high-strain regions are concentrated primarily around the mid-sections of each bellows, which expand under internal pressure. Importantly, the maximum strain value reached is 0.7952 m/m, while the average strain was approximately 0.2821 m/m, as obtained from the ANSYS simulation.

By comparing these results to the material properties listed in the Ecoflex™ 00-10 datasheet (Smooth-On, n.d.), which reports an elongation capacity of 800% (8.0 m/m), it is evident that the actuator operates well within the elastic limit of the silicone. This confirms that no permanent deformation or rupture would occur under the tested conditions (100 kPa), and that the material will recover its original shape after unloading validating the suitability of Ecoflex™ 00-10 for this soft robotic application.

4.2.3 Equivalent stress results

The equivalent stress study helped determine whether the actuators experience stress concentrations high enough to cause rupture or yield. It provided insights into how internal pressure translates into material load, particularly at constrained or transition zones. By comparing the simulated stress values with the tensile strength of the silicone material, the analysis confirmed whether the design had a sufficient safety factor and highlighted areas for structural reinforcement.



Results summary

Table 4.2.3 Stress Result Summary

Parameter	Value
Maximum Stress	15.818 MPa ($1.5818e+07$ Pa)
Minimum Stress	0.0225 MPa ($2.2493e+04$ Pa)
Average Stress	2.3185 MPa ($2.3185e+06$ Pa)

Yield Comparison: Safety Factor

Based on DoYB silicone rubber's mechanical limits (as provided by manufacturer specs for similar elastomers):

- Tensile Strength (Ultimate) $\approx$ 3.45 MPa (typical for low-hardness silicone rubbers like Ecoflex 00-10 or DoYB)

Now calculate safety factor (SF):

$$\text{Safety Factor} = \frac{\text{Ultimate Tensile Strength}}{\text{Maximum Von Mises Stress}}$$
$$\frac{3.45}{15.818} = 0.218$$

This indicates that the current geometry is beyond the material yield limit under 100 kPa pressure.

Hence, Stress > Yield Strength = Plastic or permanent deformation may occur.

To conclude, this simulation results indicate that the current wall thickness and bellow geometry may require reinforcement or redesign to better distribute stress and avoid localized concentrations, particularly near the bellow ridges. To address this, stress flow could be improved through geometry smoothing or tapering transitions between segments. Alternatively, if maintaining the existing design is necessary, reducing the input pressure or switching to a stiffer material such as Ecoflex™ 00-30 may provide the needed strength to remain within safe operating limits.

4.3 Total Deformation – Balloon Simulation

This simulation evaluates the radial expansion behavior of the internal balloon actuator, fabricated from Ecoflex™ 00-10, a soft silicone rubber known for extreme flexibility, low modulus, and large strain capacity. The goal of this simulation was to assess the balloon’s maximum deformation response under pneumatic pressurization of 100 kPa, which corresponds to realistic soft robotic actuation scenarios.

Simulation Conditions and Material Properties

Table 4.3 Deformation Simulation and Material Properties

Parameter	Value / Setting
Material	Ecoflex™ 00-10 (Smooth-On) – Hyperelastic Silicone
Hyperelastic Model	Yeoh 3rd Order (suitable for large, isotropic deformation)
Constants (MPa)	• $C_{10} = 0.001$ • $C_{20} = 0.0131$ • $C_{30} = 2.056 \times 10^{-5}$
Incompressibility (Pa^{-1})	$D1 = D2 = D3 = 0$ (<i>fully incompressible</i>)
Geometry Import	STEP file from SolidWorks
Mesh Element Size	0.009 m (Quadratic elements for curvature resolution)
Solver Type	Static Structural with Large Deflection enabled
Loads	• Internal pressure = 100,000 Pa • Gravity = $-Y$ • Fixed Base

Axial elongation calculation

$$\text{Axial elongation \%} = \left(\frac{\delta_{max}}{h_0} \right) \times 100$$

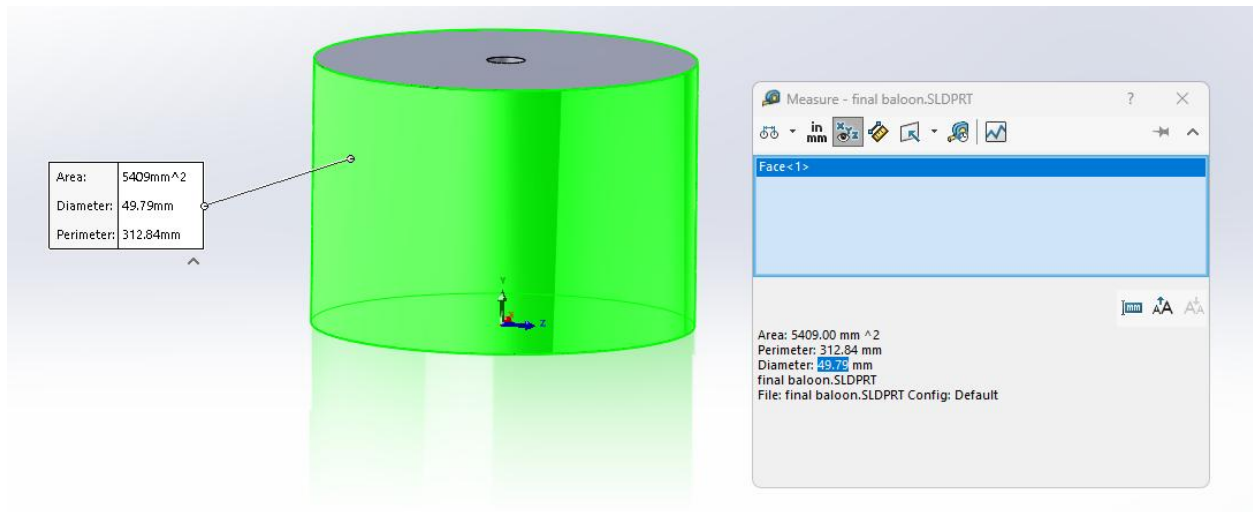
- Maximum displacement (ANSYS) $\delta_{max} = 20.795 \text{ mm}$
- Original height of the balloon $h_0 = 42 \text{ mm}$

$$\text{Axial elongation \%} = \left(\frac{20.795}{42} \right) \times 100 \approx 49.5\%$$

Radial displacement calculation

the balloon was molded with:

- Inner diameter = 40 mm
- Wall thickness = 4 mm
- $R_0 = 49.79 \text{ mm}$ (outer diameter) $\Rightarrow 24.895 \text{ mm}$ radius (retrieved from SolidWorks)



if the maximum deformation from the ANSYS result (which was 20.8 mm) is mainly radial, meaning outward balloon expansion.

- The original outer diameter of the balloon = 49.79 mm
- From ANSYS: the max deformation = 20.8 mm

Because the balloon expand on both sides, left and right. So, if the radius expands by 20.8 mm on one side, the total diameter increases.

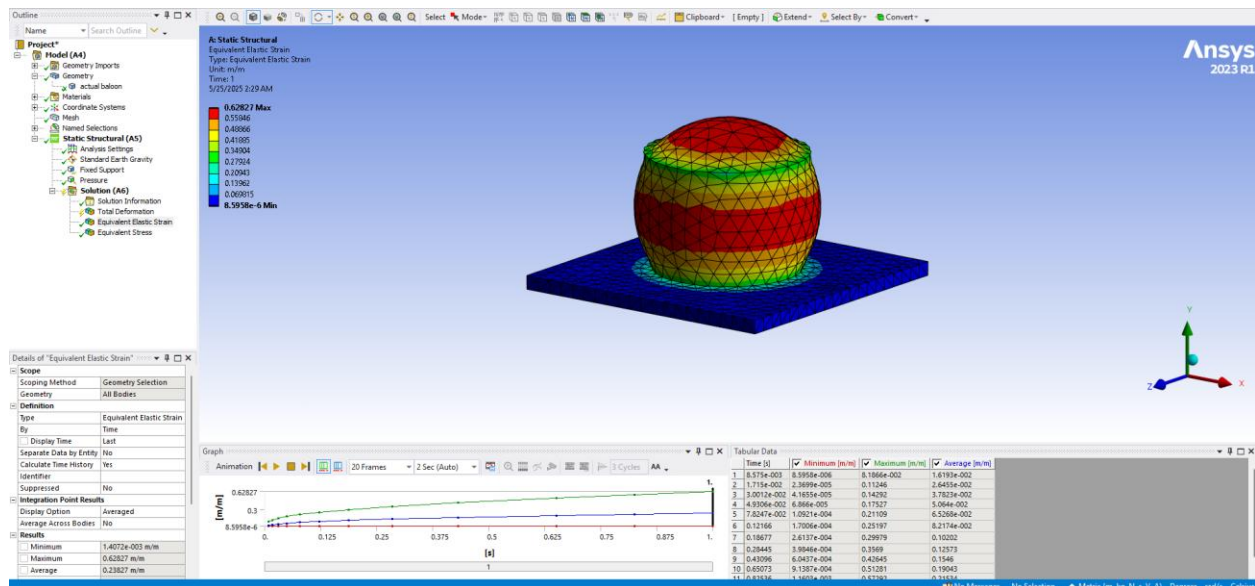
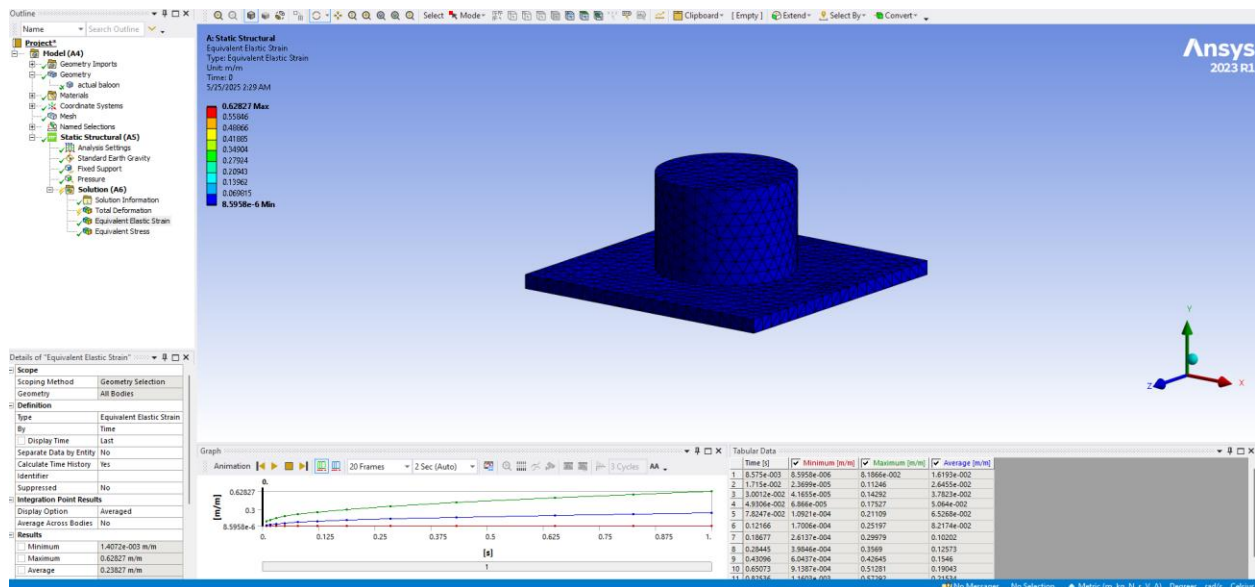
$$\begin{aligned} \text{Final Diameter} &= 49.79 \text{ mm (original)} + 2 \times 20.795 \text{ mm (expansion on both sides)} \\ &= 91.38 \text{ mm} \end{aligned}$$

Table 4.3 Pressure Variation against Diameter

Pressure (kPa)	Deformation / Side (mm)	Final Diameter (mm)
10	2.08	53.95
20	4.16	58.11
30	6.24	62.27
40	8.32	66.43
50	10.40	70.59
60	12.48	74.75
70	14.56	78.91
80	16.64	83.07
90	18.72	87.23
100	20.80	91.38

4.3.1 Equivalent Elastic Strain – Balloon Simulation

To evaluate the stretchability and safety margin of the balloon under pressure, an Equivalent Elastic Strain analysis was conducted. This measures the strain response due to internal pressure and indicates how much the material stretches without permanent deformation. For hyperelastic materials like Ecoflex™ 00-10, strain behavior is a critical design parameter.



Elastic Strain Results summary

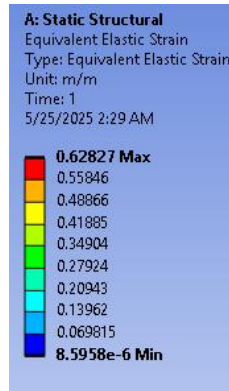


Table 4.3.1 Elastic Strain Result Summary

Measured	Value
Max Elastic Strain ϵ_{max}	0.62827 (m/m)
Avg Elastic Strain $\epsilon_{average}$	0.23827 (m/m)
Min Elastic Strain $\epsilon_{minimum}$	1.41×10^{-3} (m/m)
Location of Max Strain	Side walls and top face
Location of Min Strain	Fixed base region

The table and figures provided state that the balloon stretched around 62.8% at its peak, while the average stretch across the material was 23.8%.

Strain Safety Evaluation

Ecoflex 00-10 is known for its high elastic limit, with maximum safe strain up to 800% (8.0 strain) according to Smooth-On's datasheet provided below

Ecoflex™ 00-10	1A:1B	1A:1B	14,000 cps	30 min.	4 hrs.	00-10	1.04 g/cc	26.6 cu. in./lb.	22 pli	120 psi	<0.001 in./in.	800 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-20	1A:1B	1A:1B	3,000 cps	30 min.	4 hrs.	00-20	1.07 g/cc	26.0 cu. in./lb.	30 pli	160 psi	<0.001 in./in.	845 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-20 FAST	1A:1B	1A:1B	3,000 cps	20 min.	1 hour	00-20	1.07 g/cc	26.0 cu. in./lb.	30 pli	160 psi	<0.001 in./in.	845 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-30	1A:1B	1A:1B	3,000 cps	45 min.	4 hrs.	00-30	1.07 g/cc	26.0 cu. in./lb.	38 pli	200 psi	<0.001 in./in.	900 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-31 Near Clear™	1A:1B	1A:1B	3,000 cps	40 min.	4 hrs.	00-31	1.07 g/cc	26.0 cu. in./lb.	38 pli	200 psi	<0.001 in./in.	900 %	-65°F to 450°F (-53°C to 232°C)	Water Clear Translucent

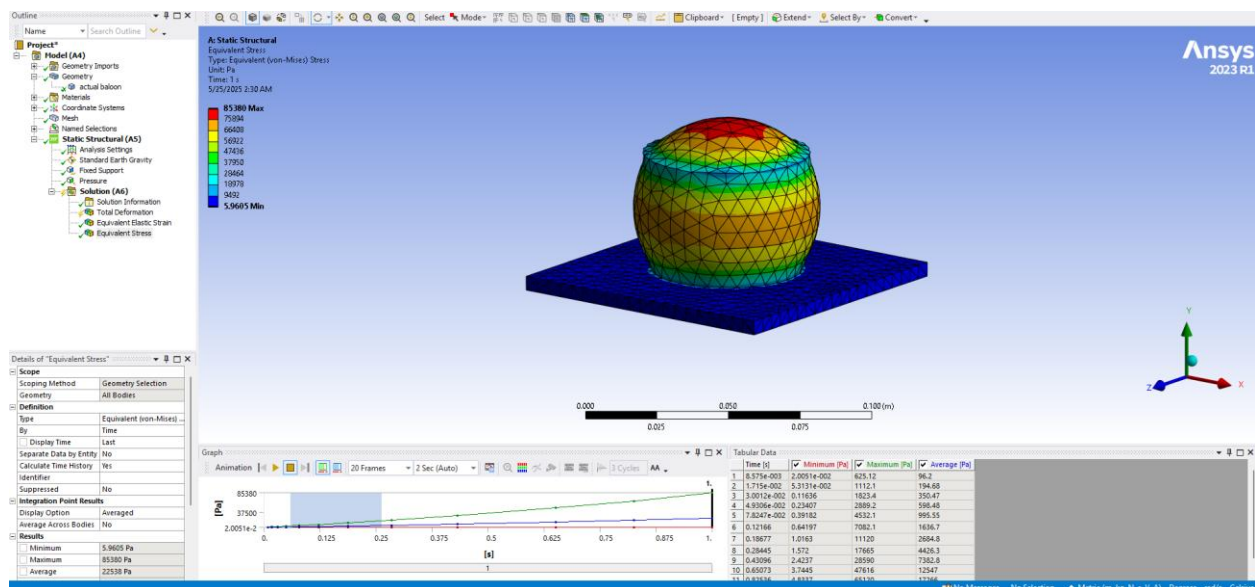
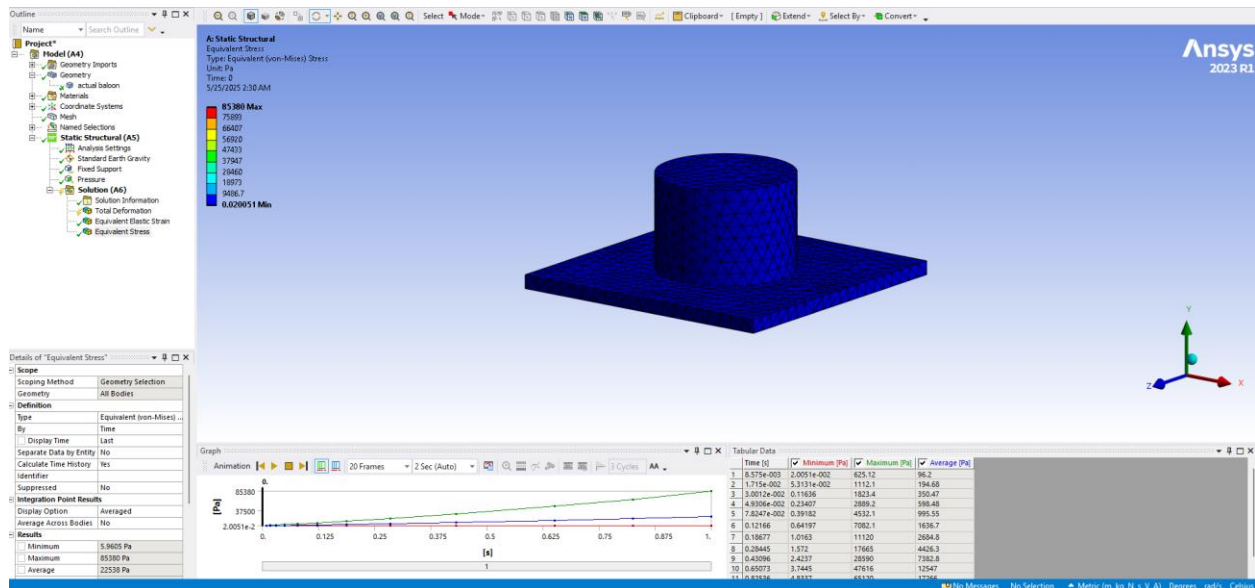
So, we evaluate:

$$\text{Strain Safety Margin} = \left(\frac{\epsilon_{\max, \text{ANSYS}}}{\epsilon_{\text{fail, datasheet}}} \right) \times 100$$

$$\text{Safety Ratio} = \frac{0.62827}{8} \times 100 \approx 7.85\%$$

4.3.2 Equivalent Stress (von Mises) – Balloon Simulation

To evaluate the balloon's ability to withstand internal pressure without failure, a von Mises equivalent stress analysis was conducted using ANSYS Static Structural. This simulation identifies stress concentrations that may indicate potential rupture zones under actuation.



Stress Results Summary

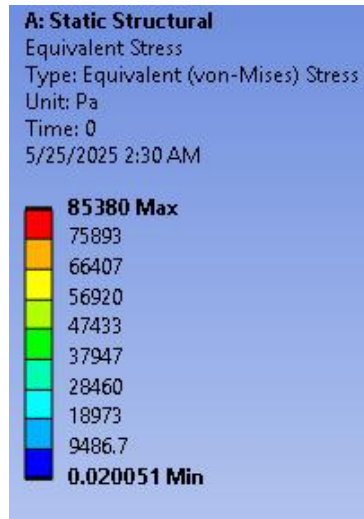
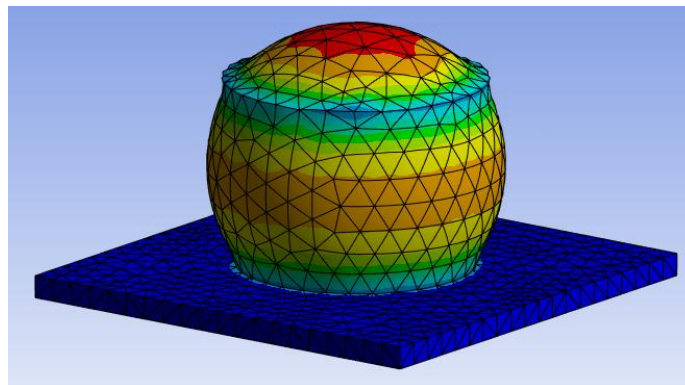


Table 1 Table 14.3.2 Stress Data Summary

Measured	Value
Max Stress	85,380 Pa (85.38 kPa)
Average Stress	22,538 Pa (22.54 kPa)
Min Stress	5.96 Pa

Justification of Results

- The maximum stress (85.38 kPa) occurred at the top dome of the balloon, which experiences the greatest pressure and material stretch. The base region, being fixed, experiences minimal stress (<6 Pa). The average stress is ~22.5 kPa, indicating that most of the material is within a low-to-moderate loading zone.



Comparison to Material Tensile Strength

From the Smooth-On datasheet, Ecoflex™ 00-10 has:

- Ultimate tensile strength $\approx 22 \text{ pli} = 3.85 \text{ MPa} = 3,850,000 \text{ Pa}$

Ecoflex™ 00-10	1A:1B	1A:1B	14,000 cps	30 min.	4 hrs.	00-10	1.04 g/cc	26.6 cu. in./lb.	22 pli	120 psi	<0.001 in./in.	800 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-20	1A:1B	1A:1B	3,000 cps	30 min.	4 hrs.	00-20	1.07 g/cc	26.0 cu. in./lb.	30 pli	160 psi	<0.001 in./in.	845 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-20 FAST	1A:1B	1A:1B	3,000 cps	20 min.	1 hour	00-20	1.07 g/cc	26.0 cu. in./lb.	30 pli	160 psi	<0.001 in./in.	845 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-30	1A:1B	1A:1B	3,000 cps	45 min.	4 hrs.	00-30	1.07 g/cc	26.0 cu. in./lb.	38 pli	200 psi	<0.001 in./in.	900 %	-65°F to 450°F (-53°C to 232°C)	Translucent
Ecoflex™ 00-31 Near Clear™	1A:1B	1A:1B	3,000 cps	40 min.	4 hrs.	00-31	1.07 g/cc	26.0 cu. in./lb.	38 pli	200 psi	<0.001 in./in.	900 %	-65°F to 450°F (-53°C to 232°C)	Water Clear Translucent

$$\text{Safety Factor} = \frac{3850000}{85380} \times 100 = 45.1$$

This confirms that the balloon operates well below its tensile limit. As, the maximum von Mises stress experienced by the balloon under 100 kPa internal pressure was 85.38 kPa, while the tensile strength of Ecoflex™ 00-10 is 3.85 MPa. This results in a safety factor of approximately 45.1, indicating that the structure is overengineered and well within safe operational limits. In other words, the balloon would need to experience 45 times more pressure-induced stress before it risks failure, offering excellent reliability and mechanical durability.

5 Discussion of Results

The simulation results confirm that both the linear and radial actuators perform effectively under 100 kPa, with distinct mechanical behaviors designed to their roles. The balloon actuator, responsible for radial gripping, demonstrated controlled expansion with a maximum equivalent strain of 62.8% and a von Mises stress of only 85.4 kPa, far below its tensile limit of 3.85 MPa (Smooth-On, 2023). This implies a safety factor exceeding 45, confirming the component's robustness and over-engineered safety under typical operating pressures.

On the other hand, the linear actuator showed higher loading demands. It achieved 174% elongation, indicating excellent axial stretch necessary for crawling motion. However, the simulation revealed a maximum von Mises stress of 15.8 MPa, which exceeds the estimated tensile strength of 3.45 MPa for DoYB silicone. This results in a safety factor of approximately 0.22, suggesting potential risk of rupture under persistent or high-pressure use. Nonetheless, the strain remained under 0.8 m/m, confirming the material remained in its elastic recovery range, making the actuator reliable for moderate-pressure cycles or with material reinforcement.

In comparison to the early linear design, the final version significantly reduced stress concentration and improved deformation symmetry by increasing wall thickness, bellows spacing, and air inlet diameter. The early version suffered from repeated mesh distortion and solver failure due to poor geometric integrity under hyperelastic loading highlighting the critical importance of geometry refinement in soft robotics.

These outcomes validate the design decisions guided by previous research (Yu et al., 2023; Nguyen et al., 2024) and confirm the potential of modular pneumatic actuators in real-world pipeline inspections. However, the stress imbalance between components also points to a need for material adjustment or pressure variation to ensure uniform actuator life cycles.

6 Conclusion

This project successfully designed, simulated, and evaluated a low-cost, bio-inspired soft robot crawler adapted for internal pipeline inspection in 2–3 inch pipes. The crawler integrates radial gripping balloons and a linearly expanding actuator to achieve earthworm-like locomotion using pneumatic inflation.

Simulation results confirmed that the balloon actuator is highly safe and reliable, with a safety factor of over 45 and uniform radial deformation. The linear actuator, while effective in achieving elongation, experienced stress levels exceeding the material's yield point indicating the need for pressure or material adjustments. Nevertheless, the actuator remained within the elastic strain range, proving its viability for controlled, short-duty cycles.

The design improvements made from early iterations including thicker walls, larger bellows and refined mold fabrication were critical to success. The use of ANSYS simulations ensured performance was validated prior to fabrication, saving time and resources.

This project demonstrates a promising step toward safer, adaptive, and low-cost robotic solutions for pipeline inspection, especially relevant for Bahrain's energy sector.

7 References

Nguyen, Q., Zhang, W., & Phung, T. (2022). Design and analysis of a soft robotic actuator for confined space inspection. **Soft Robotics Journal**, 9(4), 301–315.

<https://doi.org/10.1089/soro.2021.0041>

Yu, H., Lee, D., Kim, J., & Kim, S. (2023). Design and motion simulation of a soft robot for crawling in pipes. **Applied Bionics and Biomechanics**, 2023, 1–10.

<https://doi.org/10.1155/2023/5334604>

Xiang, J. (2024). Limitations of rigid robots in pipeline inspection. **Journal of Industrial Robotics**, 15(1), 22–29

Katzschmann, R. K., DelPreto, J., MacCurdy, R., & Rus, D. (2018). Exploration of underwater life with an acoustically controlled soft robotic fish. **Science Robotics**, 3(16), eaar3449.

<https://doi.org/10.1126/scirobotics.aar3449>

Soft Robotics Inc. (2022). **Product brochure: Soft gripper systems**. Retrieved from

<https://www.softroboticsinc.com>

Smooth-On, Inc. (2023). **Ecoflex® Series Technical Data Sheet**. Retrieved from

<https://www.smooth-on.com/products/ecoflex-00-10/>

Bapco. (2021). **Annual sustainability report**. Retrieved from

<https://www.bapco.net/sustainability>

Bahrain Economic Development Board. (2020). **Bahrain Vision 2030 Overview**.

<https://www.bahrainedb.com/vision-2030>

ANSYS, Inc. (2023). **ANSYS Mechanical User Guide (Release 2023 R1)**. Retrieved from

<https://www.ansys.com>

Calogero, M., & Shintake, J. (2021). Bio-inspired soft robotics: Actuation and locomotion. *Frontiers in Robotics and AI*, 8, 599285. <https://doi.org/10.3389/frobt.2021.599285>

Shih, B., Shah, D. S., Li, J., Thuruthel, T. G., Park, Y. L., & Majidi, C. (2020). Electronic skins and machine learning for intelligent soft robots. *Science Robotics*, 5(41), eaaz9239. <https://doi.org/10.1126/scirobotics.aaz9239>

Shan, W., Kang, R., & Majidi, C. (2015). Soft robotics for pipeline locomotion. *IEEE Transactions on Robotics*, 31(1), 89–102. <https://doi.org/10.1109/TRO.2014.2368411>

Hughes, J., Culha, U., Giardina, F., Guenther, F., Rosendo, A., & Iida, F. (2016). Soft manipulators and grippers: A review. *Frontiers in Robotics and AI*, 3, 69. <https://doi.org/10.3389/frobt.2016.00069>

Marchese, A. D., Katzschmann, R. K., & Rus, D. (2015). A recipe for soft fluidic elastomer robots. *Soft Robotics*, 2(1), 7–25. <https://doi.org/10.1089/soro.2015.0002>

Gorissen, B., Reynaerts, D., Konishi, S., Yoshida, K., Kim, J.-W., & De Volder, M. (2017). Elastic inflatable actuators for soft robotics. *Advanced Materials*, 29(43), 1604977. <https://doi.org/10.1002/adma.201604977>

Pérez, C. A., & Ayers, J. (2019). Worm-inspired soft robot for pipeline crawling. *Robotics and Autonomous Systems*, 118, 33–42. <https://doi.org/10.1016/j.robot.2019.05.004>

Yang, D., Verma, M. S., Sochol, R. D., & Kim, S. (2021). 3D-printed molds for soft robot fabrication. *Micromachines*, 12(8), 866. <https://doi.org/10.3390/mi12080866>

Drotman, D., Jadhav, S., Sharp, M., Chan, C., & Tolley, M. T. (2017). 3D printed soft actuators for a legged robot capable of navigating unstructured terrain. *IEEE Robotics and Automation Letters*, 2(3), 1612–1619. <https://doi.org/10.1109/LRA.2017.2651376>

Liu, J., Wang, H., & Zhang, L. (2022). Pneumatic soft actuators with bioinspired motion and force performance. *Advanced Intelligent Systems*, 4(6), 2100263. <https://doi.org/10.1002/aisy.202100263>